

Fear as a Barrier to Coverage: The Chilling Effect of the Public Charge Rule on Children's Medicaid Enrollment

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Abstract

Objective: To examine whether the Trump administration's 2019 public charge rule, which made Medicaid receipt a negative factor in certain immigration determinations, caused coverage losses among citizen children with a noncitizen parent, relative to those with citizen parents - also known as a "chilling effect" - even though the rule did not apply to them.

Data Sources: American Community Survey (ACS) microdata, 2014-2024 (N = 2,077,869 Medicaid-eligible U.S. citizen children below 200% of the federal poverty level); restricted-use California Health Interview Survey (CHIS) adult files, 2019-2024 (N = 220,925 adults).

Study Design: Difference-in-differences and event study models comparing citizen children in mixed-status families (at least one non-citizen parent) with children in citizen-only families; supplemental analyses on noncitizen children, naturalized children, and children with both noncitizen parents.

Principal Findings: Citizen children in mixed-status families saw a 2.0 pp decline in Medicaid enrollment after the September 2018 public charge announcement compared to children in full citizen families (SE .56, $p < 0.001$). Excluding seven states that extended Medicaid or near-equivalent state-funded coverage to undocumented adults during the study window increases the chilling estimate to -2.6 pp ($p < 0.001$). Children who were either themselves noncitizens or had two noncitizen parents saw larger effects of -2.5 to -3.8 pp ($p < 0.001$). Naturalized children and children with naturalized parents saw no detectable decline.

Conclusions: The public charge rule resulted in coverage losses among children who were not legally affected by the rule, was larger for children whose nuclear families had more noncitizens, and was concentrated in families in non-expansion states.

1. Introduction

Public charge is again an active area of immigration rulemaking. In U.S. immigration law, a “public charge” evaluation occurs when a noncitizen applies for admission to the US or for legal permanent residency (also known as a green card). If the applicant is judged likely to need government support after entry, they are deemed a “public charge” and can be denied entry or a green card status. The test has been a part of immigration law since 1882, where it was first established and used to turn away applicants for citizenship who were likely to need *cash assistance*, or “*welfare*”. While this narrow definition has continued for over the past century, regulatory actions in the last decade have expanded these determinations to include of *non-cash* benefits, such as Medicaid.

The first Trump administration’s initial 2019 expansion of the public charge rule redefined what counted as a “public charge” for immigration purposes to include receipt of non-cash benefits like Medicaid, Supplemental Nutrition Assistance Program (SNAP), and federal housing assistance. While the rule applied only to certain noncitizens applying for admission or adjustment of status and not their children or family, the opaque and punitive nature of the regulation created uncertainty among who would be affected - and may have caused entire households to avoid public benefits. This fear and confusion-driven avoidance among people a policy does not legally reach is commonly called a “chilling effect”.

Using the historical 2019 case, I evaluate the central empirical question of this paper: If such fear circulates through immigrant communities, does it translate into measurable coverage losses among the populations the rule never touched? I focus on U.S. citizen children in mixed-status families - U.S.-citizen children who live with at least one noncitizen parent - and who are eligible for Medicaid on exactly the same terms as any other citizen children. As the rule altered no income threshold, categorical eligibility rule, or benefit package, coverage changes for these children should plausibly only occur through parental or household behavior; as such I am able to utilize national survey data to provide evidence on chilling effects and use state survey data to supplement and elaborate on effects of the fear across different immigrant populations.

The primary analysis uses microdata from the American Community Survey (ACS), the Census Bureau's large annual household survey, on Medicaid-eligible U.S. citizen children from 2014 to 2024 to estimate difference-in-differences and event-study models comparing children in mixed-status families with those in citizen-only families. The ACS provides a nationally representative sample large enough to detect statistically meaningful coverage changes, as well as individual-level data on household immigration composition and coverage spanning the full public charge policy cycle from pre-announcement through the Medicaid unwinding; for purposes of this paper, I define unwinding as the period (2023-2024) when states resumed eligibility checks and disenrollments after the pandemic-era Medicaid continuous enrollment requirement ended.

This question is a matter of substance for many reasons, first and foremost related to consistency of children's healthcare coverage. Medicaid is the primary source of health coverage for low-income children in the United States - in 2026, Medicaid and the Children's Health Insurance Program (CHIP) covers approximately 35 million children, nearly half of all children below 200% of the federal poverty level (FPL), and decades of research establish that childhood Medicaid coverage improves health outcomes, educational attainment, and long-run economic self-sufficiency (Currie and Gruber, 1996; Cohodes et al., 2016; Wherry and Meyer, 2016; Miller and Wherry, 2019). Any policy that reduces consistency of enrollment could potentially have consequences that compound over a child's lifetime. This work sits in three related areas of literature, first and foremost being evidence on the 2019 public charge rule itself. The closest study that provides direct public charge evidence is Barofsky et al. (2020), who used county-level administrative enrollment data through June 2019 and estimated approximately 260,000 fewer children were enrolled in Medicaid after the September 2018 announcement. Their county-exposure design treats noncitizen population share as treatment intensity, which mixes true exposure with the state policy environments, sanctuary infrastructure (local government policies that limit cooperation with federal immigration enforcement), and urban composition of high-noncitizen-share counties. In addition, their treatment window closes before implementation, reversal, and the Medicaid unwinding.

Whether the decline reversed once the rule was withdrawn, and what happened when the unwinding began, are unanswered by this study.

Second, the welfare reform policy from the late 1990s establishes a similar pattern of chilling a generation earlier. The Personal Responsibility and Work Opportunity Reconciliation Act of 1996 (PRWORA) barred most legal immigrants who arrived after its enactment from federal means-tested benefits for five years and left states to decide whether to cover others with state funds. Its passage was followed by benefit declines that reached immigrants the law did not restrict: Kandula et al. (2004) estimated a 3 percentage point Medicaid decline among immigrants explicitly exempt from PRWORA's restrictions, and Kaushal and Kaestner (2005) found larger effects - 10–13% - on uninsurance.

Third and last is immigration enforcement, meaning arrests, detentions, and deportations of noncitizens, which reduces safety-net participation among citizens and noncitizens alike without any change in eligibility rules. Watson (2014) finds that the decline in Medicaid participation among children of noncitizens around welfare reform is largely explained by a contemporaneous spike in federal immigration enforcement, and Alsan and Yang (2024) showed that Secure Communities, a 2008-2014 program that checked the fingerprints of everyone booked into local jails against immigration records, reduced participation in safety-net programs, including SNAP and Affordable Care Act coverage, among Hispanic citizen households. Their results are most consistent with fear transmitted through social networks, meaning households responded to enforcement that relatives and neighbors experienced rather than to their own contact with it. Twersky (2022) reports lower Medicaid/CHIP take-up among citizen children in immigrant families in states with restrictive immigration laws, and Kravitz et al. (2024) report lower Medicaid and SNAP enrollment among eligible adults where ICE detainer requests were more common.

I add to the literature in several ways. First, the paper provides individual-level quasi-experimental evidence consistent with a chilling effect of the public charge rule on Medicaid enrollment. In addition, I examine whether the pattern correlates with household exposure to immigration risk, and through which mechanisms. My paper also builds on the concept of administrative burden, which distinguishes the

learning, compliance, and psychological costs of interacting with government programs (Herd and Moynihan, 2018). The announcement contributed to all three domains of administrative burden at once: it created a regulatory framework that even some attorneys had difficulty parsing, signaled that enrollment systems would scrutinize immigrant applicants more closely, and tied benefit use to immigration risk. This set of “public charge” policies reversed the previous welcoming of eligible groups into enrollment in public health insurance: where the Affordable Care Act insurance coverage expansion signaled that public coverage was welcome and pulled in people who had been eligible all along, the public charge change was a negative information shock about whether participation was safe for families whose eligibility never changed.

2. Background

2.1 History of public charge doctrine

In U.S. immigration law, “public charge” is a ground of inadmissibility - a reason the federal government may deny a noncitizen entry to the country or, more commonly, deny their application to adjust status to lawful permanent resident (LPR, or “green card”). U.S. citizens are never subject to such a determination and legal permanent residents are rarely, if ever, affected as well. The legal test is forward-looking: an adjudicating officer at U.S. Citizenship and Immigration Services (USCIS) or Department of State consular assesses whether, in their judgment, the noncitizen is “likely to become a public charge” - or in other words, be a substantive burden to the public system. In practice, a determination is dependent on a number of criteria: (i) what categories of benefit use count as evidence of likely dependence, (ii) which other personal characteristics (age, income, family ties, English proficiency) are weighed, and (iii) how heavily benefit use is weighted relative to those other factors.

The first known formal instance of public charge is the Immigration Act of 1882, which barred admission to any person who would likely be dependent on government cash subsistence (Capps, Greenberg, Fix, and Zong, 2018). This law was clarified under DHS’ 1999 Field Guidance, which noted that public

charge determinations should be based on utilization of cash-based assistance programs like SSI and that most non-cash programs, like Medicaid, should not be considered.

The 2019 final rule specified that receipt of Medicaid (for non-pregnant adults under age 65), SNAP, or federal housing assistance for more than 12 months in any 36-month period would be a “heavily weighted negative factor” in public charge determinations under a “totality of the circumstances” test (84 Fed. Reg. 41,292). While *how* a public charge determination was made changed, it did not change who could be subject to one. Those who can are noncitizens applying for admission or to adjust to lawful permanent resident status, most of them family-based applicants; refugees, asylees, and a number of other humanitarian categories are exempt by statute. Everyone else was outside the rule: U.S. citizens were still not subject to public charge determinations, lawful permanent residents were only at risk at narrow re-entry margins and not for renewal of LPR status, and naturalized citizens are not currently subject to the doctrine in any form.

A rule can change behavior among people it does not directly apply to by raising perceived participation costs. Take-up among programs falls when those costs rise, even where formal eligibility is unchanged (Moffitt, 1983; Currie, 2006; Herd and Moynihan, 2018; Bhargava and Manoli, 2015). The 2019 rule was interpreted far more broadly than its legal reach, which was reasonable given how the rule was described and how little was settled about its final form: surveys found that many immigrants, including naturalized citizens and U.S. citizens in mixed-status families, believed their own or a family member’s benefit use could carry consequences (Bernstein, McTarnaghan, and Gonzalez, 2019; Gonzalez, Bernstein, and Karpman, 2020). Of importance for this paper is that the rule explicitly excluded children’s Medicaid and CHIP from consideration, but due to incomplete information and the child-parent linked nature of these benefits, many families were unsure who exactly would be affected.

2.2 Timeline of recent public charge related events

Table 1. Public charge chronology and related coverage policy, 2017-2026

Date	Event
January 2017	Draft executive order on public benefits leaks

October 2017	DHS places the rule on the Unified Regulatory Agenda
September 2018	Proposed rule released; published October 10, 2018 (83 Fed. Reg. 51,114)
August 14, 2019	Final rule published (84 Fed. Reg. 41,292)
October 2019	Federal courts in several circuits enjoin implementation
January 2020	Supreme Court stays most injunctions; a Seventh Circuit exception leaves Illinois enjoined
February 24, 2020	Rule takes effect in most states
March 2020	Federal Medicaid continuous enrollment condition begins during the public health emergency (FFCRA §6008(b)(3))
November 2, 2020	Northern District of Illinois enters a Rule 54(b) judgment vacating the rule
March 9, 2021	USCIS stops applying the 2019 rule; DHS removes the vacated text and restores the pre-2019 regulations (86 Fed. Reg. 14,221)
December 23, 2022	Replacement final rule takes effect (87 Fed. Reg. 55,472, published September 9, 2022)
March 31, 2023	Continuous Medicaid enrollment condition ends; the Medicaid unwinding begins
September 18, 2026	Scheduled effective date of the 2026 final rule rescinding the 2022 framework (91 Fed. Reg. 45,324)

Notes: Dates are taken from the Federal Register documents and public laws cited in the second column. The 2026 rule had not taken effect as of the date of this analysis.

Table 1 above outlines recent events surrounding the public charge rule. I note the importance of several events even before the final rule was released in 2019, as early events started the informational cascade that could produce chilling effects months before and in concert with a formal rule change. The September 2018 notice of proposed rulemaking generated extensive coverage in both English- and Spanish-language media, frequently describing the rule in broad terms without fully conveying the details, including exemptions. Community-based organizations worked to spread accurate information, but the rule's complexity and uncertainty about its final form during the 14-month comment period created an informational environment in which fear was both rational (the rule did apply to some immigrants and did involve Medicaid) and overinclusive (many families believed they were affected when they were not). For purposes of my analysis, I also note that the long announcement period and litigation cycle muddles interpretation because it creates a long period of uncertainty with no clear pre/post period for policy treatment assignment. Even after injunctions, families could not be confident that the rule would not ultimately take effect - and even after the Biden administration reversal of the

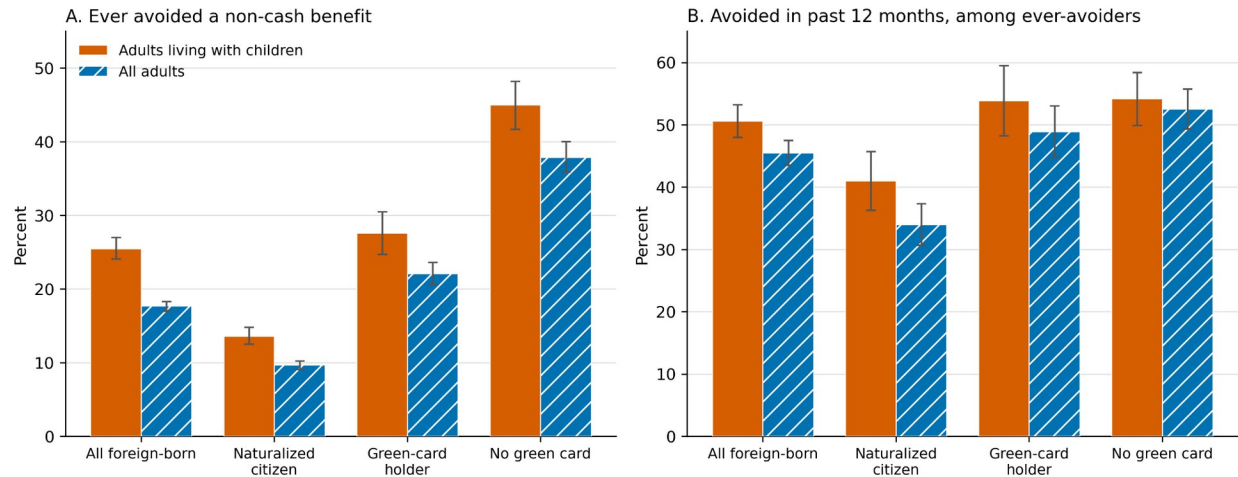
policy, the memory of the policy and possibility of its revival may have altered immigrant behavior and kept some families from re-enrolling in the longer term.

2.3 Reported benefit avoidance among California immigrant families

The chilling mechanism rests on families actually avoiding benefits out of immigration concerns, so before turning to the national data I first document and confirm that they report doing so. The California Health Interview Survey (CHIS), the largest state health survey in the country, asks foreign-born adults about this directly, and I report its descriptive shares here as background for the analysis that follows. CHIS is administered annually by the UCLA Center for Health Policy Research, samples roughly 20,000-25,000 adults per cycle, and identifies U.S.-born citizens, naturalized citizens, noncitizens with green cards, and noncitizens without green cards; it is used in this paper only for these descriptive shares, computed on the restricted-use 2019-2024 adult files.

Beginning in 2019, CHIS asked foreign-born adults whether they had ever avoided applying for a non-cash government benefit because they worried it would hurt their own or a family member's chances of obtaining a green card or U.S. citizenship. Figure 1 reports survey-weighted shares for adults living with a child under 18 and for all adults, pooled over 2019-2024, and Table O1 reports the underlying estimates. I am unable to conduct a formal California analysis that mirrors my main ACS estimations due to power issues, as well as California's uniquely immigrant-protective policies, which make it a less useful case study for national implications.

Figure 1. CHIS Adults Fear-based benefit avoidance by documentation group, 2019-2024



Notes: Panel A plots the share who ever decided to avoid applying for a non-cash benefit over immigration concerns, and Panel B plots the share who avoided applying in the past 12 months among those who ever avoided. Bars are survey-weighted shares, pooled over the 2019-2024 CHIS adult files, with 95% confidence intervals from jackknife replicate-weight standard errors. Both items are asked only of foreign-born respondents, and the Panel A item is retrospective and unbounded in time. Table O1 reports the shares, standard errors, and unweighted denominators. Author's calculations from restricted-use CHIS data accessed through a data use agreement with the UCLA Data Access Center.

Avoidance is common: a quarter of immigrant adults living with children reported having declined a non-cash benefit over immigration concerns at some point, and among those, roughly half did so within the past year. The avoidance rates were highest where legal exposure is most direct, with 45.0% of adults without green cards who live with children avoiding public benefits versus 27.6% of green-card holders. Interestingly, even among naturalized citizens, 13.6% avoided benefit enrollment, even though they cannot formally be the subject of a public charge determination. I use this descriptive analysis as motivation and proof for my central chilling effect thesis: immigration-related benefit avoidance extends past legally exposed groups.

3. Data

3.1 American Community Survey

The national analysis uses ACS microdata accessed through IPUMS USA for survey years 2014 through 2024. The ACS is the largest household survey in the United States outside the decennial census, sampling approximately 3.5 million addresses per year and providing nationally representative individual-level data on demographics, household composition, income, health insurance coverage, and citizenship

status. The ACS identifies whether each individual is a U.S. citizen by birth, citizen by naturalization, or noncitizen. It does not distinguish among noncitizen types - lawful permanent residents, temporary visa holders, and undocumented immigrants. This limitation is common to all major federal surveys, and it means the mixed-status family definition necessarily pools families with different types of noncitizen parents.

3.2 Sample

The main sample includes U.S. citizen children ages 0 to 18 in households below 200% of FPL. This FPL threshold approximates the upper bound of Medicaid/CHIP eligibility in most states; while exact thresholds vary by state, child age, and program (some states extend CHIP eligibility to 300% FPL or higher), 200% FPL cutoff captures the overwhelming majority of the population for whom public coverage is a realistic option.

The full citizen-child sample contains 6,482,027 observations across eleven survey years (2014-2024). The preferred below-200% FPL sample contains 2,077,869 observations: 389,899 (18.8%) in mixed-status families and 1,687,970 (81.2%) in citizen-only families. The two added pre-announcement years contribute 615,565 children in 2014 and 610,140 in 2015, which extends the pre-period available for parallel-trends testing from two survey years to four. I present full-sample estimates for all income levels ($N = 6,482,027$) as secondary specifications (Appendix G). Because 2014 and 2015 fall inside the ACA's coverage-expansion rollout, Appendix C discusses how that overlap affects the interpretation of the extended pre-period leads.

3.3 Exposure

I classify children as living in mixed-status families if at least one identified parent in the household roster is a noncitizen, while citizen-only families are those in which all identified parents are U.S. citizens.

While I use this as my primary definition and primary treated population, I separately further decompose mixed-status families and examine if children with one noncitizen parent see less chilling than two noncitizen parents are the most exposed configuration; these results are reported in Tables 8 and 10. The

child must be a U.S. citizen in both groups. Children whose parents cannot be identified through the IPUMS household-roster pointers (~4% of the eligible-age citizen-child sample) are excluded. This classification captures the household-level immigration exposure central to the chilling-effect mechanism.

3.4 Outcomes

The primary outcome is current Medicaid enrollment, with secondary outcomes including any public coverage, any health insurance coverage, and uninsurance. The distinction between Medicaid and any public coverage is of note because some children may shift between programs (e.g. Medicaid to CHIP) in response to the public charge environment. The uninsurance measure captures the most consequential outcome (children with no coverage at all), but may be noisier if some adjustment involves shifts to employer-sponsored or individual-market insurance.

3.5 Estimation conventions and data quality

All models are weighted least squares using IPUMS person weights, with state and year fixed effect - specifications that add finer sets of fixed effects say so where they appear - and controls for child age dummies, sex, and race and ethnicity, and standard errors clustered on state. The Census Bureau released only experimental one-year estimates for 2020 and instructs that “[d]ata users should NOT compare 2020 ACS 1-year experimental estimates with any other data.” As such, the specification excluding 2020 is co-primary and is reported alongside the baseline throughout, with 2020-based estimates labeled experimental wherever they appear. Table 2 reports descriptive statistics for the two groups.

Table 2. ACS Children Descriptive statistics, with household incomes below 200% FPL, 2014-2022

Variable	Mixed-status families	Citizen-parent families
N, pre-proposal (2014-2018)	212,597	915,494
N, post-proposal (2019-2022)	120,521	592,746
Child age (mean, pre-period)	8.11	8.55
Two-parent family (share, pre-period)	0.775	0.477
Hispanic (share, pre-period)	0.814	0.222
Non-Hispanic White (share, pre-period)	0.063	0.443

Non-Hispanic Black (share, pre-period)	0.051	0.243
Asian and Pacific Islander (share, pre-period)	0.053	0.023
Income-to-poverty decile within sample (mean, pre-period)	4.21	4.29
Medicaid/CHIP, pre-period weighted mean	0.797	0.706
Uninsured, pre-period weighted mean	0.078	0.054
Any coverage, pre-period weighted mean	0.922	0.947

Notes: U.S.-citizen children ages 0 to 18 below 200% FPL with at least one identified resident parent. Demographic entries are pooled sample shares and means; coverage entries are survey-weighted means over 2014-2018. Author's calculations (IPUMS USA, 2014-2022).

4. Empirical Approach

4.1 Primary comparison

The primary contrast is between two groups of U.S.-citizen children with household incomes below 200% FPL: those living with at least one identified resident noncitizen parent, and those whose identified resident parents are all citizens. The pooled version of the model is a two-group difference-in-differences (DID) specification on repeated cross-sections:

$$Y_{ist} = \alpha \text{Mixed}_i + \beta(\text{Mixed}_i \times \text{Post}_t) + \delta_s + \lambda_t + X_i \gamma + \varepsilon_{ist},$$

where Y_{ist} = ACS-reported Medicaid or other means-tested public coverage for child i in state s in survey year t , Mixed_i = an indicator for living with at least one identified resident noncitizen parent, Post_t = an indicator for survey years 2019 and later, δ_s and λ_t = state and year fixed effects, X_i = individual controls (age dummies, sex, race and ethnicity), and ε_{ist} represents an error term. The coefficient of interest is β , the differential change for citizen children in mixed-status families relative to those in citizen-parent families. I avoid “treated” and “untreated” labels because the paper has no single well-dated intervention and because chilling is a spillover that can potentially reach the comparison group as well.

4.2 Timing and specifications

My primary design is a two-way fixed effects (TWFE) pooled DID: Post equals one from 2019 onward, and β gives the average differential over 2019-2022. I use this design as primary because the setting is a single national policy event with two comparison groups rather than a staggered rollout across states. Every family faced the same announcement at the same time, so the classic two-group DID applies and the heterogeneity-robust estimators built for staggered adoption are not needed; I also utilize doubly robust estimators below as estimator checks. I note since there is uncertainty in policy treatment timing due to multiple potential “triggering” events - the 2017 leak, 2018 proposed rule, and 2019 finalized rule, the actual effective policy date, the vacating of the rule - I run models with earlier treatment timing or transition coefficients during middle years as well. The pooled estimation window is therefore 2014-2022. The 2023-2024 survey years enter only the event studies and the descriptive splits, where they are labeled as unwinding years, because the Medicaid unwinding changed enrollment for both groups in ways a pooled post indicator would absorb.

I run a co-primary event study design, taking 2017 as the reference year, 2018 as its own transition coefficient, and the following as post, and designate it as the “policy-period convention” in my subsequent tables. Each convention is estimated in three ways: the baseline TWFE model with state and year fixed effects and the Section 3.5 controls; the same model excluding the experimental 2020 ACS, its companion specification throughout; and a variant with state-by-year fixed effects, which absorbs every state-specific annual shock. As an estimator check, I re-estimate both conventions with the doubly robust DID estimator of Sant’Anna and Zhao (2020). One further specification gives race, ethnicity, income position (the child’s income-to-poverty decile within the sample), age, and family structure (single- versus two-parent families and the presence of other adult relatives) their own year effects.

4.3 Complementary comparisons

Two additional designs address specific threats on the same ACS child sample.

4.3.1 Long-naturalized-parent comparison

Citizen-parent families differ from mixed-status families in immigrant background as well as citizenship. I specify an alternative model that compares children of noncitizen parents with children of immigrant parents who naturalized in 2013 or earlier as the control, excluding families whose parents naturalized in 2019 or later so that group membership is not set after the announcement. Both groups are restricted to families in which every immigrant parent arrived by 2013. I report four weighting schemes: survey weights only, entropy balancing solved on 2014-2018 (Hainmueller, 2012), propensity-based overlap weights with the score frozen on the same years, and a common-support restriction (Appendix D).

4.3.2 Exposure and dose decomposition

I estimate two mutually exclusive multi-group models, one for two-parent families and one for single-parent families, so that each parental configuration is compared with families of the same structure. Three contrasts carry the main evidence because each holds family structure and the other parent's nativity fixed: a noncitizen against a naturalized parent alongside a U.S.-born parent, the same contrast alongside a naturalized parent, and a single noncitizen against a single naturalized parent. In the single-parent model the omitted reference is a single U.S.-born parent, so the single-noncitizen row itself is the comparison with a native-born citizen parent, and the contrast with a naturalized parent is reported alongside it. Each is also estimated with parents' origin region, arrival cohort, and English proficiency given their own year effects.

4.4 Immigrant-inclusive states

Seven jurisdictions extended Medicaid or near-equivalent state-funded coverage to undocumented adults during the study window (California, Illinois, Oregon, New York, Colorado, Washington, and the District

of Columbia); I refer to these as the immigrant-inclusive states and run additional event study models with a dummy variable indicating if a state had noncitizen child expansion at that time.

4.5 Estimation and inference

Standard errors are clustered on state. I report wild cluster bootstrap p-values and bootstrap-t confidence intervals for the main and alternative comparison-group estimates (Cameron, Gelbach, and Miller, 2008). The parent-pair contrasts come from three separate regressions, so they get a joint wild cluster bootstrap that applies the same cluster draws to all three models (Appendix E), and the entropy-balanced design utilizes a state-cluster bootstrap.

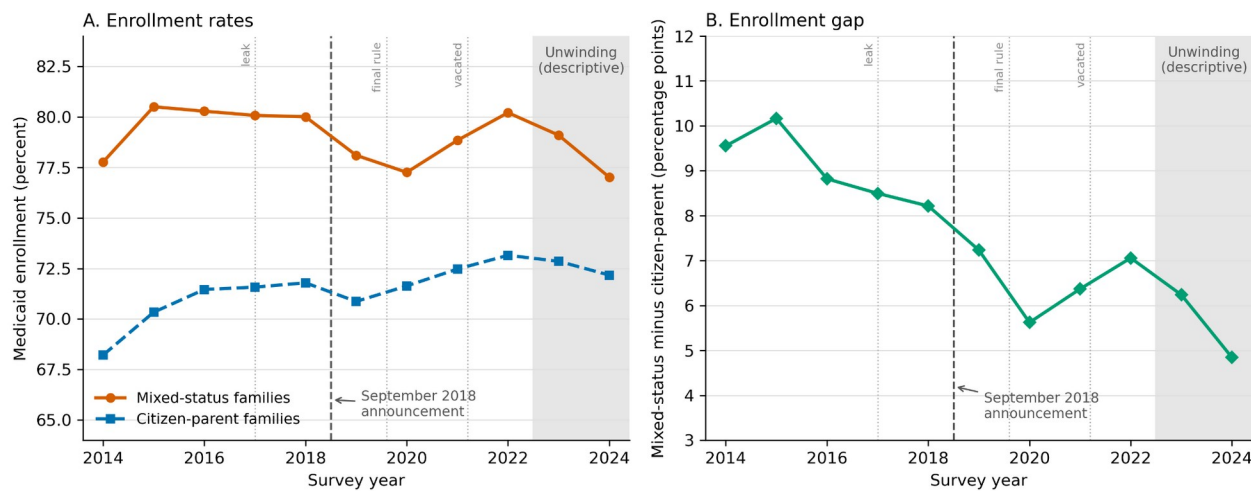
For pre-trend sensitivity, I follow Rambachan and Roth (2023) and compute the largest post-proposal violation of parallel trends. Where a coefficient is not statistically distinguishable from zero I report its confidence interval, and minimum detectable effects (MDEs) at 80% power appear in table notes.

5. Results

5.1 Raw trends

Figure 2 plots weighted Medicaid/CHIP coverage for the two groups, along with the gap between them.

Figure 2. Raw Medicaid/CHIP coverage trends, U.S.-citizen children with household incomes below 200% FPL, 2014-2024



Notes: Panel A plots survey-weighted rates for the two groups; note the truncated vertical axis. Panel B plots the difference between the series in percentage points. The dashed line marks the September 2018 proposal, which falls between the 2018 and 2019 survey panels, and the shaded region covers the 2023-2024 Medicaid unwinding years. Lighter dotted lines mark the January 2017 draft-order leak, the August 2019 final rule, and the March 2021 vacatur (Table 1). Author's calculations (IPUMS USA, 2014-2024).

By-year trends in coverage are fairly parallel in the pre-treatment period. Mixed-status families have approximately 9% higher Medicaid/CHIP coverage than all-citizen families in the pre period; after public charge changes, mixed-status coverage drops and narrows the gap to 7.5% in 2019 and 5.8% in 2020, and did not recover subsequently. Mixed-status children had higher coverage throughout, about 78 to 80% against 68 to 73%, as their household incomes are lower in the income distribution within the low-income band (Table 2). Both groups had increased enrollment from 2020-2022, with mixed-status families seeing larger gains, presumably due to the Medicaid pause on unwindings due to the COVID-19 pandemic; the larger gains are lost during the subsequent Medicaid unwinding period, suggesting more mixed-status families gained coverage that they were eventually ineligible for compared to citizen families.

5.2 Main estimates

Table 3 reports the pooled and phase estimates under the three specifications, Figure 3 reports the co-primary event study, and Table 4 compares estimators.

Coverage among mixed-status children fell relative to citizen-parent children after the proposal. The pooled differential over 2019-2022 is -2.00 points (SE 0.56, $p < 0.001$), or -1.67 in the companion specification excluding the experimental 2020 file. This translates to a relative decline of approximately 3% among children whose own eligibility rules did not change. The estimate is essentially unchanged under state-by-year fixed effects.

Table 3. ACS Children Change in Medicaid/CHIP coverage over time, main estimates

Panel A. Pooled convention (post = 2019-2022, reference period 2014-2018)

Specification	All years	Excluding 2020 ACS
Baseline	-2.00*** (0.56)	-1.67*** (0.44)
State-by-year fixed effects	-2.00*** (0.43)	-1.52*** (0.38)

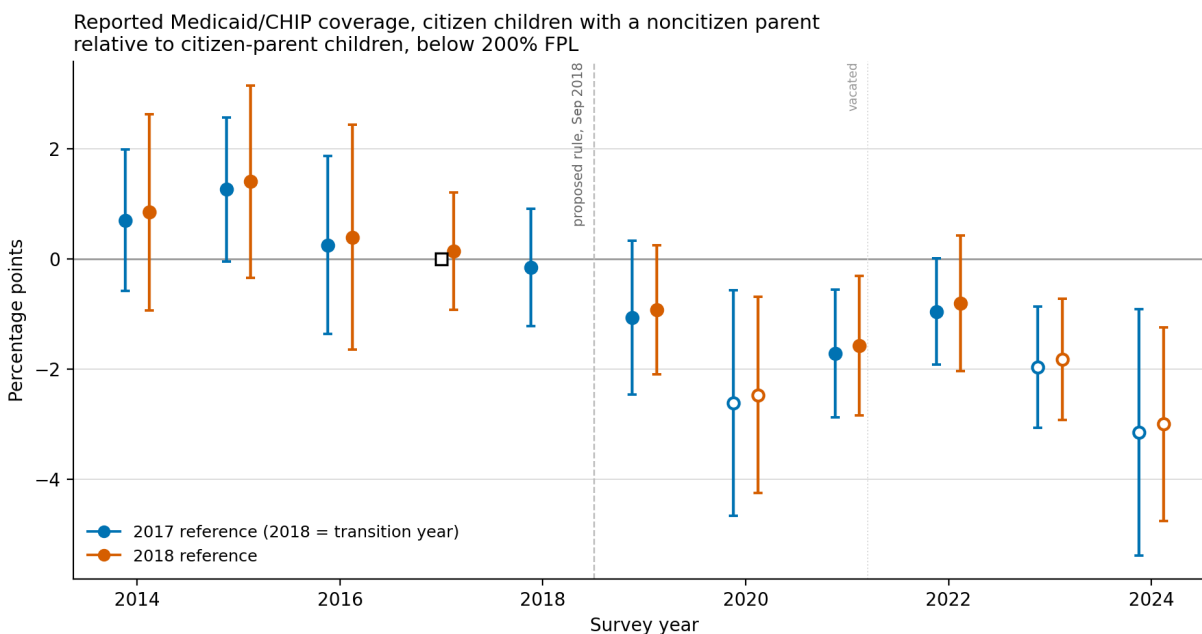
Pre-period mean, mixed-status (%)	79.7	79.7
Pre-period mean, citizen-parent (%)	70.6	70.6
Observations	1,841,358	1,697,491
Clusters (states)	51	51

Panel B. policy-period convention (post = 2019-2022, reference period 2014-2017, transition coefficient 2018)

Specification	2018	2019	2020	2021-2022	N
Baseline	-0.72 (0.76)	-1.63* (0.91)	-3.17** (1.25)	-1.91*** (0.45)	1,841,358
Excluding 2020 ACS	-0.73 (0.76)	-1.62* (0.92)	-	-1.91*** (0.45)	1,697,491
State-by-year fixed effects	-0.67 (0.67)	-1.49** (0.73)	-3.60*** (1.01)	-1.75*** (0.43)	1,841,358

Notes: Each cell is the coefficient on the mixed-status-by-period interaction from a weighted least squares regression of reported Medicaid/CHIP coverage on state and year fixed effects (state-by-year where stated) and the Section 3.5 controls, below 200% FPL, 2014-2022, with standard errors clustered on state in parentheses. Panel A pools 2019-2022 with 2018 in the reference period; Panel B gives 2018 its own transition coefficient. The 2020 estimates rest on the experimental 2020 ACS, and the excluding-2020 variant is the baseline's companion specification. Pre-period means are survey-weighted Medicaid/CHIP rates over the Panel A reference years, 2014-2018; over the 2014-2017 reference period of Panel B they are 79.6% and 70.3%. Each Panel B row is a single regression, so its sample and its 51 state clusters are constant across the phase columns. Wild cluster bootstrap p-values for the Panel A baseline (Rademacher weights on the 51 state clusters, unrestricted and recentred, 9,999 replications) are 0.011 with all years and 0.004 excluding 2020; this bootstrap is the governing inference for the baseline row, and the significance markers reflect the analytic p-values. *p<0.10, **p<0.05, ***p<0.01. Author's calculations (IPUMS USA, 2014-2022).

Figure 3. Event study, reported Medicaid/CHIP coverage, U.S.-citizen children below 200% FPL



Hollow markers: the 2020 experimental ACS and the 2023-24 unwinding years; the 2023-24 years are excluded from every pooled estimate. Whiskers are 95% confidence intervals from state-clustered standard errors.

Notes: Event studies interacting the mixed-status indicator with survey-year indicators, estimated over 2014-2024 with state and year fixed effects, the Section 3.5 controls, and standard errors clustered on state. The blue series uses 2017 as the omitted reference and gives 2018 its own transition coefficient, and the orange series uses a 2018 reference. Hollow markers denote the experimental 2020 ACS and the 2023-2024 unwinding years. The dashed line marks the September 2018 proposal, the design's treatment basis, and the lighter dotted line marks the March 2021 vacatur. Author's calculations (IPUMS USA, 2014-2024).

Three leads before the 2017 reference are individually insignificant and the 2018 transition coefficient is -0.15 points (SE 0.53, $p = 0.78$), which is suggestive that chilling did not occur until actual enactment; dropping 2018 slightly increases the pooled estimate to -2.14 ($p = 0.003$). The phase coefficients rise and fall: -1.63 points in 2019, the proposal-to-finalization year ($p = 0.082$); -3.17 in 2020, the implementation year, resting on the experimental file; and -1.91 ($p < 0.001$) in 2021-2022, after the rule stopped being applied and while continuous enrollment kept most existing enrollees covered. A joint test that the three phases are equal gives $p = 0.094$, and dropping both 2018 and the experimental 2020 file leaves 2019 at -1.62 and 2021-2022 at -1.91, with an equality test of $p = 0.700$. The 2022 coefficient on its own is negative under either reference year without being statistically established. In short, the phase most directly tied to the policy signal is the least precisely estimated, and the most precisely estimated phase is the hardest to attribute to the rule.

Table 4. ACS Children’s Change in Medicaid/CHIP coverage over time, TWFE and doubly robust estimators

Panel A. Pooled convention (post = 2019-2022, reference period 2014-2018)

Estimator	All years	Excluding 2020 ACS
TWFE	-2.00*** (0.56)	-1.67*** (0.44)
Doubly robust, imputation	-0.98* (0.52)	-0.65 (0.48)
Doubly robust, IPW	-0.96** (0.48)	-0.64 (0.46)
Pre-period mean, mixed-status (%)	79.7	79.7
Pre-period mean, citizen-parent (%)	70.6	70.6
Observations	1,841,358	1,697,491
Clusters (states)	51	51

Panel B. Policy-period convention (ost = 2019-2022, reference period 2014-2017, transition coefficient 2018)

Estimator	2018	2019	2020	2021-2022
TWFE	-0.72 (0.76)	-1.63* (0.91)	-3.17** (1.25)	-1.91*** (0.45)
Doubly robust, imputation	-0.36 (0.80)	-0.51 (0.75)	-2.09** (0.93)	-0.82 (0.52)

Doubly robust, IPW	-0.36 (0.78)	-0.49 (0.72)	-2.07** (0.86)	-0.80 (0.50)
Observations, doubly robust	1,128,091	1,111,186	1,063,058	1,296,596

Notes: Panel conventions and sample as in Table 3; standard errors clustered on state (51 clusters) in parentheses, and pre-period means as defined there. TWFE rows repeat the Table 3 baseline. Doubly robust rows implement the Sant'Anna and Zhao (2020) repeated-cross-section estimators, with the Section 3.5 controls entering both the outcome model and the group-membership model; the imputation and IPW variants differ in how the outcome regression is used. Each Panel B doubly robust estimate compares the 2014-2017 reference years with the named year alone, which is why its sample varies by column; the TWFE phase coefficients come from one regression on the full sample of 1,841,358. *p<0.10, **p<0.05, ***p<0.01. Author's calculations (IPUMS USA, 2014-2022).

The doubly robust estimates are smaller in magnitude than TWFE models but largely similar to the main estimates. Pooled estimates are ~ -0.97 ($p = 0.05$) about half the baseline and ~ -0.65 excluding 2020; by phase, only the 2020 coefficient is distinguishable from zero ($p = 0.016$ and 0.024).

5.3 Coverage categories

Table 5 breaks the estimate into coverage categories and income bands. Public-only coverage falls 2.14 points and private-only coverage rises 1.51 points, so the private increase offsets much of the public decline in the aggregate. Dual coverage, the category that would move if children were adding private coverage on top of public coverage, is flat ($p = 0.56$).

Table 5. ACS Children Change in coverage over time, outcomes and income bands

Panel A. Primary outcome and income bands

Sample	Outcome	Estimate (pp)	SE	p	95% CI	N
Below 200% FPL	Medicaid/CHIP	-2.00	0.56	<0.001	[-3.13, -0.88]	1,841,358
Below 200% FPL	Any public coverage	-2.03	0.49	<0.001	[-3.02, -1.04]	1,841,358
Below 200% FPL	Private coverage	+1.65	0.29	<0.001	[+1.07, +2.22]	1,841,358
Below 200% FPL	Uninsured	+0.47	0.48	0.335	[-0.50, +1.43]	1,841,358
Below 300% FPL	Medicaid/CHIP	-2.41	0.59	<0.001	[-3.60, -1.23]	2,804,622
Below 138% FPL	Medicaid/CHIP	-1.55	0.54	0.006	[-2.64, -0.45]	1,171,175
Below 100% FPL	Medicaid/CHIP	-1.42	0.53	0.009	[-2.47, -0.36]	761,251

Panel B. Mutually exclusive coverage categories, below 200% FPL

Category	Estimate (pp)	SE	p	Excluding 2020	Mixed-status mean (%)	Citizen-parent mean (%)
Public coverage only	-2.14	0.42	<0.001	-1.79	75.87	64.72
Private coverage only	+1.51	0.21	<0.001	+1.28	11.99	23.10
Both public and private	+0.14	0.23	0.555	+0.12	3.40	6.21
Uninsured	+0.47	0.48	0.335	+0.35	8.10	5.55
Other public only (Medicare or veterans' coverage)	+0.03	0.09	0.762	+0.04	0.64	0.41

Notes: "Medicaid/CHIP" abbreviates ACS-reported Medicaid or other means-tested public coverage throughout the paper. Each row is the mixed-status by post interaction from a separate regression with state and year fixed effects, the Section 3.5 controls, and standard errors clustered on state; post is survey year 2019 or later and the pooled window is 2014-2022. Panel B categories are built so that every child-year falls in exactly one category and the five coefficients sum to zero. The 80%-power minimum detectable effect is 1.37 points for uninsurance. Author's calculations (IPUMS USA, 2014-2022).

Uninsurance shows no detectable change: +0.47 points with a 95% confidence interval of [-0.50, +1.43] but fails its joint pre-trend test ($p = 0.018$). Private coverage has significant increases, potentially indicating shifting to this coverage, although the repeated cross-sectional nature of the ACS makes this difficult to prove. Estimates stay negative in every income band and are smallest in the lowest-income samples, the group least able to substitute private coverage; the bands do not isolate where eligibility limits matter.

5.4 Long-naturalized families as control

As citizen-parent families differ from mixed-status families in immigrant background, I estimate alternative models with long-naturalized parents (2013 or earlier) as the control as they may be more demographically similar. Table 6 presents these results.

Table 6. ACS Children Change in Medicaid/CHIP coverage over time, long-naturalized comparison

Specification	Group or contrast	Estimate (pp)	SE	p	Excluding 2020	N
Established-immigrant	Noncitizen vs	-2.42	0.69	<0.001	-2.08	394,553

design, unbalanced	long-naturalized parent					
Established-immigrant design, entropy-balanced	Noncitizen vs long-naturalized parent	-1.57	1.72	0.366	+0.02	394,553
Established-immigrant design, overlap-weighted	Noncitizen vs long-naturalized parent	-0.72	0.68	0.298	+0.01	394,553
Established-immigrant design, common support	Noncitizen vs long-naturalized parent	-1.28	0.63	0.046	-0.73	231,608

Notes: Both groups are restricted to families in which every immigrant parent arrived by 2013 and comparison parents naturalized by 2013. Appendix D diagnoses the weighting schemes and reports the four-level parental-citizenship decomposition. Confidence intervals are [-5.03, +1.89] (entropy-balanced), [-2.09, +0.65] (overlap-weighted), and [-2.53, -0.02] (common support). Weighted least squares, state and year fixed effects, standard errors clustered on state, 2014-2022. Author's calculations.

Long-naturalized families seem to have slightly lower, albeit mostly nonsignificant, reductions in coverage. These differences become even less pronounced in entropy-balanced, overlap-weighted, and common support models.

5.5 Robustness

Table 7. ACS Children Change in Medicaid/CHIP coverage over time, robustness by family

Family and specification	Estimate (pp)	SE	p	N
Data quality				
Baseline, 2014-2022	-2.00	0.56	<0.001	1,841,358
Excluding the experimental 2020 ACS	-1.67	0.44	<0.001	1,697,491
Excluding records with an allocation flag	-1.65	0.54	0.004	1,512,407
U.S.-born citizen children only	-1.91	0.59	0.002	1,804,558
Composition				
Composition-reweighted to the pre-period distribution	-1.80	0.64	0.007	1,841,358
Same, adding FPL band to the reweighting cells	-1.20	0.60	0.050	1,841,358
Same, adding FPL decile to the reweighting cells	-1.04	0.61	0.095	1,841,358
Post-2018 naturalizers dropped from the comparison group	-2.00	0.57	<0.001	1,832,638
Geography				
PUMA-by-year fixed effects	-2.36	0.51	<0.001	1,841,350
Placebo: citizen-only families, high vs low	-0.13	0.63	0.836	1,508,240

noncitizen-share states				
Group-specific trends				
Race-, income-decile-, age-, and family-structure- by-year fixed effects	-0.59	0.47	0.211	1,841,328

Notes: Each row is the mixed-status by post interaction from a separate regression over 2014-2022, with state and year fixed effects except where a finer set is stated, and standard errors clustered on state. The composition-reweighted rows hold the mixed-status group at its pooled 2014-2018 distribution of parent origin, education, arrival cohort, child age, and family structure. PUMA-by-year specifications are within-year comparisons because PUMA boundaries were redrawn for the 2020 census. The placebo row compares citizen-only families in states above versus below the median noncitizen population share. Appendices G and H report the full income-conditioning and trend-adjustment ladders. Author's calculations.

Table 7 reports further robustness checks. Excluding every record with an allocation flag gives -1.65 points, standardizing to U.S.-born children gives -1.91, reweighting to the pre-period composition gives -1.80, or -1.04 to -1.20 once income cells enter the reweighting, and dropping post-2018 naturalizers changes nothing. Geography is not the problem either: PUMA-by-year fixed effects give -2.36, and citizen-only families in high-noncitizen-share states show no post-2018 change.

When race, ethnicity, income position, age, and family structure are each given their own year effects, the estimate falls to -0.59 points and loses statistical significance (SE 0.47, $p = 0.211$); the baseline's 95% confidence interval excludes this point estimate, so the two specifications produce estimates that are statistically distinct from each other .

5.6 Exposure gradient

If the response runs through immigration exposure, the decline should be larger where exposure is greater. Table 8 decomposes the mixed-status category by parental configuration.

Table 8. ACS Children Change in Medicaid/CHIP coverage over time, parental configuration

Group or contrast	Estimate (pp)	SE	p	Joint pre-trend p
Two-parent: one U.S.-born, one naturalized (BZ)	+2.37	1.20	0.054	-
Two-parent: two naturalized (ZZ)	+2.17	0.95	0.027	-
Two-parent: one U.S.-born, one noncitizen (BN)	-1.28	0.91	0.165	-
Two-parent: one naturalized, one noncitizen (ZN)	-0.46	0.81	0.575	-
Two-parent: two noncitizen (NN)	-2.14	0.84	0.014	-

Single naturalized parent	+1.40	0.66	0.039	-
Single noncitizen parent	-1.83	0.48	<0.001	-
Contrast: one U.S.-born and one noncitizen parent vs one U.S.-born and one naturalized parent (BN vs BZ)	-3.65	1.36	0.010	0.436
Contrast: one naturalized and one noncitizen parent vs two naturalized parents (ZN vs ZZ)	-2.63	0.88	0.004	0.585
Contrast: single noncitizen parent vs single naturalized parent (single N vs single Z)	-3.23	0.85	<0.001	0.254

Notes: B denotes a U.S.-born citizen parent, Z a naturalized citizen parent, and N a noncitizen parent. Two mutually exclusive multi-group regressions, one for two-parent families with two U.S.-born parents as the reference (N = 1,065,250) and one for single-parent families with a single U.S.-born parent as the reference (N = 776,108). Contrasts are linear combinations of the two group-by-post coefficients with state-clustered standard errors, and joint pre-trend p is the Wald test on the contrast's pre-proposal leads. Appendix E reports the wider set of contrasts, their joint bootstrap inference, and the family-unit exposure ladder. Author's calculations (IPUMS USA, 2014-2022).

Decline rises with parental noncitizenship, within two-parent families and among single-parent families alike. Single-parent rows are measured against a single U.S.-born parent, so the -1.83 point estimate for single noncitizen parents is the direct comparison with a native-born citizen parent. The three contrasts that hold family structure and the other parent's nativity fixed are all negative, and all three pass their joint pre-trend tests; their joint wild cluster bootstrap p-values are 0.013 (BN vs BZ), 0.003 (ZN vs ZZ), and below 0.001 (single-parent contrast).

As parent-pair contrasts still compare households that differ in origin, time in the country, and language, I estimate further models with origin, arrival cohort, and parental English proficiency year effects and report them below.

Table 9. ACS Children Change in Medicaid/CHIP coverage over time, parent-pair contrasts under added year interactions

Added fixed effects	U.S.-born pair (BN vs BZ)	Naturalized pair (ZN vs ZZ)	Single parent (N vs Z)
Base: state and year FE	-3.73 (1.34), p = 0.007	-2.54 (0.87), p = 0.005	-3.36 (0.85), p < 0.001
Parent origin region-by-year	-3.09 (1.61), p = 0.061	-1.73 (0.80), p = 0.035	-2.77 (0.85), p = 0.002
Parent arrival cohort-by-year	-3.28 (1.42), p = 0.025	-2.97 (0.93), p = 0.002	-3.01 (0.83), p < 0.001
Parent English proficiency-by-year	-3.01 (1.30), p = 0.024	-2.51 (0.82), p = 0.004	-2.59 (0.87), p = 0.004
Origin- and arrival-cohort-by-year	-2.59 (1.79), p =	-2.03 (0.84), p = 0.019	-2.16 (0.84), p =

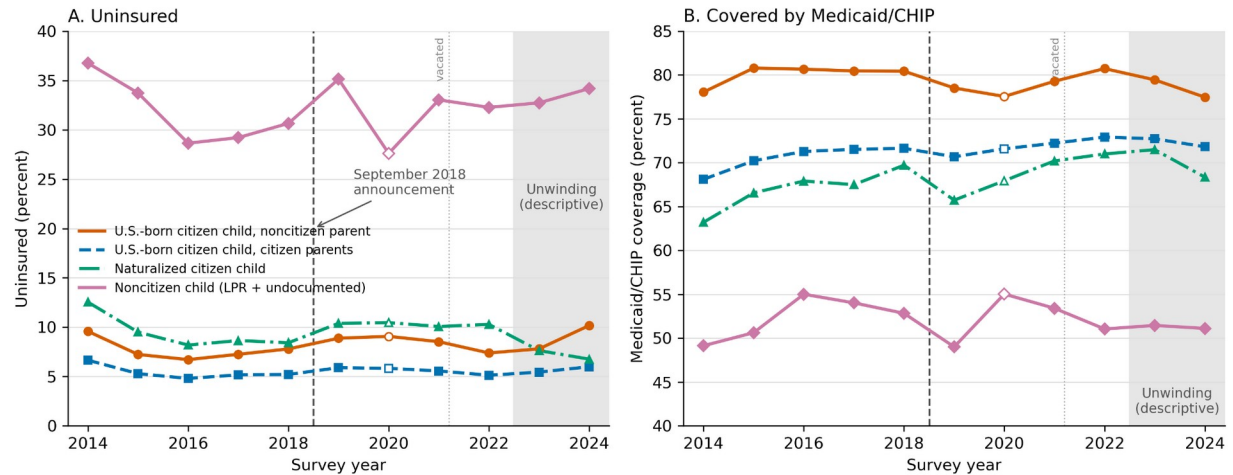
	0.155		0.013
Origin-, arrival-cohort- and English-proficiency-by-year	-2.41 (1.77), p = 0.181	-1.99 (0.82), p = 0.019	-1.87 (0.87), p = 0.036

Notes: Columns name each contrast by the nativity of the other parent: a noncitizen against a naturalized parent alongside a U.S.-born parent, the same contrast alongside a naturalized parent, and a single noncitizen against a single naturalized parent. Standard errors clustered on state in parentheses. Each column is estimated on the two contrasted configurations alone with the naturalized side as the omitted reference, so no reference group is spanned by the added fixed effects; Appendix E reports the same variants in the parameterization of Table 8. Origin region and arrival cohort come from the resident parents' birthplace and year of immigration, and English proficiency is the most proficient English-speaking ability among the resident parents (ACS SPEAKENG). Every joint pre-trend test passes, minimum p 0.106. The adjusted specifications place larger same-signed 2016 and 2017 leads than the base. For the naturalized pair those leads move from -0.74 and -1.54 points to -2.62 and -2.02 under the origin-and-cohort adjustment, and for the single-parent contrast from -0.89 and -0.72 to -2.93 and -2.52. Author's calculations.

Under the joint origin-, arrival-cohort-, and English-proficiency-by-year effects, the naturalized-pair and single-parent contrasts remain negative and significant at the 5% level, retaining 56 to 80% of their base magnitude. Children whose parent is a noncitizen therefore lost relatively more coverage than children whose parent is naturalized, even within origin region, arrival cohort, parental English proficiency, state, and year. The U.S.-born-pair contrast keeps its sign and 65 to 70% of its magnitude, and its interval widens to include zero; it is the smallest of the three pairs and the one whose sides differ most in origin composition, so the 80%-power minimum detectable effect rises from 3.82 to 5.13 points.

Figure 4 plots weighted uninsurance and Medicaid/CHIP coverage over 2014-2024 for low-income children grouped by their own citizenship status, with U.S.-born citizen children split by parental citizenship. Noncitizen children, a group that pools lawful permanent residents and undocumented children, have the lowest Medicaid/CHIP coverage and by far the highest uninsurance in every year, at around 28 to 37% uninsured against under 13% for every citizen group. Naturalized citizen children sit in between on Medicaid/CHIP coverage, and U.S.-born children with citizen parents have the lowest uninsurance throughout.

Figure 4. ACS Children, Uninsurance and Medicaid/CHIP coverage by child citizenship status, below 200% FPL, 2014-2024



Notes: Survey-weighted rates for children ages 0 to 18 below 200% FPL with at least one identified resident parent, grouped by the child's own citizenship status, with U.S.-born citizen children split by parental citizenship; children born abroad to U.S.-citizen parents are excluded. The noncitizen group pools lawful permanent residents and undocumented children. The dashed line marks the September 2018 proposal, and the lighter dotted line marks the March 2021 vacatur. The series span 2014-2024; hollow markers denote the experimental 2020 ACS, and the shaded region covers the 2023-2024 unwinding years. Author's calculations (IPUMS USA, 2014-2024).

Table 10 turns this gradient into estimates. Each row re-estimates the pooled and phase models with the mixed-status treatment group replaced, in turn, by U.S.-born citizen children with two noncitizen parents, by noncitizen children, and by naturalized citizen children, holding the comparison group at the paper's citizen-parent control; in each design every other exposed group is removed from the estimation sample rather than left to sit in the comparison group. Citizen children with two noncitizen parents, the most exposed parental configuration in Table 8, decline 2.49 points (SE 0.75, $p = 0.002$), modestly more than the baseline 2.00. Noncitizen children decline 2.20 points (SE 0.84, $p = 0.011$), strengthen to -3.09 without the experimental 2020 file, and post the largest 2019 coefficient of any design in the paper, -3.79 points ($p = 0.007$). Naturalized citizen children show no detectable change: +0.42 points (SE 1.43, $p = 0.769$) on a clean pre-period ($p = 0.710$).

Table 10. ACS Children Change in Medicaid/CHIP coverage over time, alternative treatment groups

Treatment group	Pooled (SE)	p	Excluding 2020	2019	2020	2021-2022	Joint pre-trend p	N
U.S.-born citizen child, two noncitizen parents	-2.49 (0.75)	0.002	-2.18	-1.89	-3.58	-2.53	0.006	927,644

Noncitizen child	-2.20 (0.84)	0.011	-3.09	-3.79	+0.75	-2.91	0.032	1,558,367
Naturalized citizen child	+0.42 (1.43)	0.769	+0.73	-0.77	-0.18	+1.89	0.710	1,514,150

Notes: Each row is a separate two-group design in which the named group replaces the mixed-status treatment group and the comparison group is the paper's control, citizen children whose identified resident parents are all citizens, over the pooled 2014-2022 window with the Section 3.5 controls, state and year fixed effects, and standard errors clustered on state. Exclusions by row: the first excludes citizen children with exactly one noncitizen parent and all foreign-born children, and restricts both groups to two-resident-parent families; the second excludes naturalized children and every citizen child with a noncitizen parent; the third excludes noncitizen children and every non-naturalized citizen child with a noncitizen parent. Phase columns come from one regression per row with 2014-2017 as the pre-period and separate indicators for 2018, 2019, 2020, and 2021-2022; the 2018 transition coefficients are -0.65, -0.78, and +1.59 points, none significant. The joint pre-trend column is the Wald test on the 2014-2016 leads relative to the 2017 reference in the Figure Q1 event studies. The 80%-power minimum detectable effect for the naturalized-child row is 4.01 points. Treated pre-period coverage rates are 83.1%, 52.3%, and 67.1%. The 2020 columns rest on the experimental ACS file. Author's calculations from American Community Survey microdata (IPUMS USA, 2014-2024).

I note two caveats to the findings above. The two-noncitizen-parent and noncitizen-child designs fail pre-trend testing. Noncitizen children are also themselves exposed and face different eligibility rules, including the five-year bar and state coverage regardless of status, so their estimate may mix chilling with direct exposure.

5.7 State policy environment

Excluding immigrant-inclusive states increases the chilling effect from approximately 2% to 2.3%, while limiting the analysis to just non-expansion states results in a 2.6% reduction in Medicaid enrollment. I note significant pre-trends issues with the non-expansion and expansion-only models and flag the results as descriptive.

Table 11. ACS Children Change in Medicaid/CHIP coverage over time, state policy environment

Specification	Term	Estimate (pp)	SE	p	N
Excluding CA, IL, OR, NY, CO, WA, DC	Mixed-status by post	-2.24	0.77	0.006	1,364,624
Excluding IL, WA, MN, NY, CA (pre-proposal classifier)	Mixed-status by post	-2.29	0.73	0.003	1,387,697
National interaction, pre-proposal classifier	Mixed-status by post, non-protective states	-2.33	0.73	0.003	1,841,358
National interaction, pre-proposal classifier	Mixed-status by post by protective state	+0.56	0.85	0.509	1,841,358
National interaction, adult-coverage classifier	Mixed-status by post by protective state	+0.51	0.91	0.578	1,841,358

Non-expansion states only	Mixed-status by post	-2.60	0.80	0.005	734,775
Expansion states only	Mixed-status by post	-1.70	0.39	<0.001	1,106,583

Notes: The first row drops the seven jurisdictions that extended Medicaid or near-equivalent state-funded coverage to undocumented adults during the window. The second uses a classifier fixed before the proposal, whether the state already covered income-eligible children regardless of immigration status as of September 2018. The interaction rows keep all states and interact the mixed-status by post term with a protective-state indicator; their 80%-power minimum detectable effects are 2.43 and 2.61 points. Expansion status is measured on July 1, 2019 (KFF, 2026). Author's calculations.

6. Discussion

6.1 Main interpretation

I find that citizen children in mixed-status families saw a 2.0 percentage point decline in Medicaid enrollment after the September 2018 public charge announcement. The decline was larger outside the states with immigrant-inclusive policies and larger the more noncitizen parents a child had, and it was largest among children who were themselves noncitizens. Naturalized citizens did not see decreases in coverage. A decline of 2.0 points - or a relative decline of about 3 percent among children whose own eligibility never changed - may seem small against a coverage rate near 80 percent. However, on the roughly 5.7 million citizen children in mixed-status families below 200% FPL in 2018, it corresponds to about 115,000 fewer children enrolled in a given post-announcement year, with losses falling on the group least able to replace Medicaid with private coverage. It is also the same order of magnitude as the welfare-reform chilling estimates of the late 1990s and paints a similar picture: children legally entitled to Medicaid lost coverage because their parents faced a difficult choice between covering the family and avoiding a program they feared could carry immigration consequences.

However, compositional differences in groups may account for the reductions in coverage, as indicated by doubly-robust estimation and models with additional controls. Many findings are not necessarily robust to a large battery of alternative specifications; I argue this is due to the aforementioned compositional differences, multi-phased nature of the rule, confounding factors like COVID-19, and difficulty in accurate treatment assignment due to “chilling” conceptually affecting a group that is not technically affected.

By-year phase estimates also provide mixed, uncertain results. The 2019 phase, which should hypothetically be the most direct signal period, is imprecise at -1.63 points ($p = 0.082$). The 2020 phase carries the largest coefficient, rests on the experimental file, and is likely contaminated by COVID-19 effects. The 2021-2022 phase is the most precisely estimated, but it falls after the rule stopped being applied, in years when continuous enrollment kept most existing enrollees covered.

6.2 Relation to prior literature

The magnitude of the main estimates are broadly consistent with prior work. Barofsky et al. (2020) estimated effects in roughly the 2-4 percentage point range for the most exposed counties, implying approximately 260,000 fewer children enrolled. My estimate is somewhat smaller as an absolute count, but the designs are not directly comparable. In that sense, I interpret my result less as a contradiction and more as a targeted estimate of the child-level margin.

The welfare reform precedent is also informative. Kandula et al. (2004) estimated a 3 percentage point decline after PRWORA, a magnitude comparable to the estimates here. Kaushal and Kaestner (2005) found larger effects on uninsurance, while Borjas (2003) found that the immigrants targeted by the federal Medicaid cutbacks raised their labor supply and gained employer coverage, so their overall coverage did not fall. That Medicaid-to-private substitution is the same pattern that appears in Section 5.3 here, where private coverage rose as Medicaid fell and uninsurance did not detectably change. The parallel suggests that chilling effects of this size are a recurring feature of immigration-health policy interaction, not something unique to the 2019 rule. The enforcement literature provides additional context: Watson's (2014) finding that the decline in Medicaid participation among children of noncitizens around welfare reform is largely explained by enforcement shows how much the policy environment, as distinct from formal eligibility rules, shapes enrollment decisions. Alsan and Yang's (2024) Secure Communities estimates showed similarly large spillovers to citizen households, with effects most consistent with network transmission of fear transmitted through social networks, where households that never had contact with enforcement responded to what happened to relatives and neighbors.

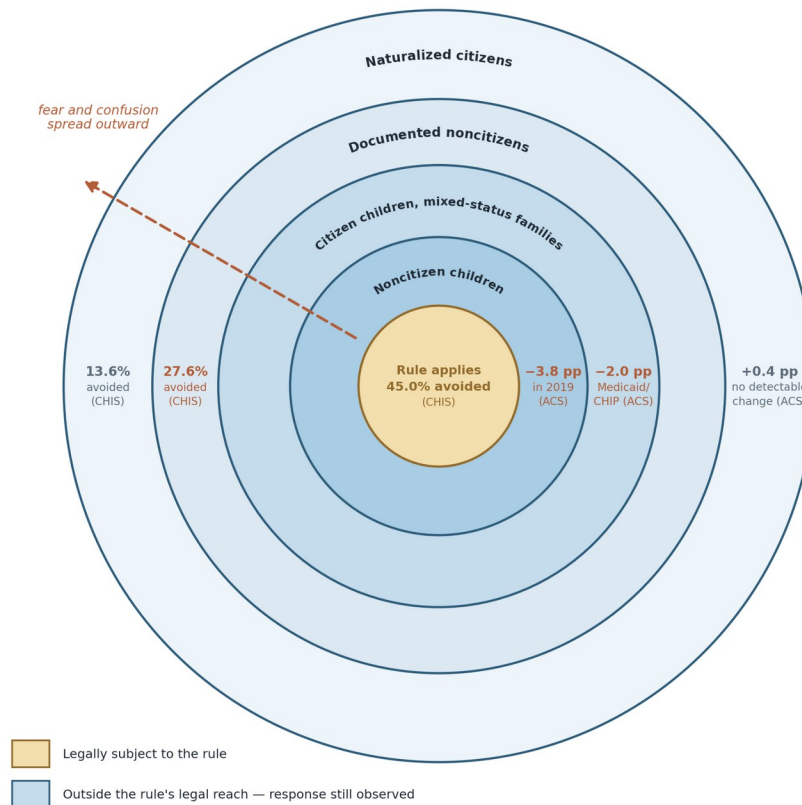
6.3 Mechanisms

While my research design cannot directly test specific mechanisms, I argue that the pattern of my results is most consistent with the following conclusion: the public charge rule generated fear that spread through immigrant communities via media coverage, word of mouth, and community-based organization warnings. That fear raised the perceived cost of enrolling in Medicaid for families who had, or believed they had, members vulnerable to immigration consequences. The fear was overinclusive: it reached populations (citizen children, naturalized citizens) that the rule did not formally target, because the rule's complexity made it hard for families to assess their actual risk. The fear interacted with a pre-existing administrative environment in which immigrants face systematically greater documentation demands when seeking coverage, and the combination of elevated fear and heightened bureaucratic scrutiny created a deterrent sufficient to reduce enrollment among eligible populations.

The policy implication is less about getting the design of the rule right than about expecting spillovers of this kind. The chilling response was foreseeable: reducing immigrants' reliance on public benefits was the rule's stated purpose, and the rule's own regulatory impact analysis anticipated disenrollment, including by U.S. citizens in mixed-status households (84 Fed. Reg. 41,292). The coverage losses among citizen children are therefore better read as a predictable consequence of the policy than as a design flaw in it. Spillovers like this are common when immigration rules touch programs that families share, and their health consequences fall on people the rules never named. Where the goal is to undo them, outreach aimed only at adjustment-of-status applicants will under-correct, and reassurance has to reach green-card holders and naturalized citizens with mixed-status relatives as well.

I use Figure 5 below to argue that my results indicate that chilling effects are stronger the closer a family is to the truly affected cohort (non-citizens applying for green cards).

Figure 5. Chilling Magnitudes by Proximity to populations legally subject to the rule



Notes: Visualization is not to any scale; pictured as above only to show fading chilling effects as individuals have better citizenship status. The center is the population legally subject to the 2019 rule, noncitizens seeking lawful permanent residence, whose own benefit use could be counted against them in a determination, the rule's full legal scope; 45.0% of California immigrant adults without green cards who live with children report having ever declined a non-cash benefit over immigration concerns (Figure 1 and Table O1). Moving outward: noncitizen children declined 3.79 points in 2019 ($p = 0.007$), the proposal-to-finalization year. Their own Medicaid was excluded by the rule's under-21 carve-out, but their own immigration status was at stake, so this group's estimate mixes chilling with direct exposure; the ring reports the 2019 phase coefficient rather than the pooled estimate because 2019 is the phase most directly tied to the policy signal (Table 7). U.S.-citizen children in mixed-status families, whose Medicaid was explicitly excluded from the rule, declined 2.00 points on the pooled 2014-2022 window (Table 3). Green-card holders and other documented noncitizens, ordinarily not subject to a new public charge determination for benefit use, report a 27.6% ever-avoidance share (Figure 1 and Table O1). Naturalized citizens are the only group fully outside the rule: naturalized citizen children show no detectable coverage change, +0.42 points (Table 7), and naturalized adults report the lowest avoidance share, 13.6% (Table O1). ACS estimates are author's calculations from American Community Survey microdata (IPUMS USA, 2014-2024); CHIS shares are author's calculations from restricted-use CHIS data accessed through the UCLA Data Access Center, pooled over the 2019-2024 adult files, for adults living with children.

Said another way, the chilling response traveled through information channels rather than legal ones.

Word of the rule moved through immigrant networks, media coverage, and community organization communications faster than any individualized legal analysis could, which would explain why avoidance appears in groups the rule could not legally reach. The same channels plausibly ran slower in reverse, since a family that heard the rule was coming also had to hear that it was gone, and the design does not test either step directly.

6.4 Policy implications

6.4.1 Immigration policy as health policy

Perhaps the most fundamental implication of this paper is that immigration policy can function as de facto health policy. A rule directed at immigration adjudication produced measurable coverage losses among U.S. citizen children who were never legally subject to it; as such, future cost-benefit analysis for similar immigration rules should include health coverage effects as foreseeable consequences, not unanticipated side effects.

6.4.2 Administrative environment and enrollment infrastructure

The documentation-demand gradient shows that enrollment systems treat immigrants substantively differently from native-born citizens. This administrative gap creates the conditions under which fear can translate into coverage loss. If enrollment systems were designed to minimize documentation demands for eligible populations (for example, by not asking parents about their own immigration status when processing children's Medicaid applications), the chilling effect of policies like the public charge rule might be significantly reduced. California's relatively protective policy environment muting the direct adult coverage response makes the administrative-vulnerability even more important for understanding why even protective states remain susceptible to chilling.

6.4.3 Persistence of fear

Reversing a restrictive and/or punitive rule does not quickly erase its effects, especially when subsequent administrative shocks interact with already-elevated fear. The persistence through reversal and deepening during the unwinding suggest that policymakers should anticipate lasting behavioral consequences from restrictive immigration policies. Outreach, community education, and proactive re-enrollment efforts may be necessary to counteract residual effects of fear-generating policies, even after those policies are formally rescinded - these repeated efforts perhaps allow for the "slow melting" that is required to reverse the chilling.

6.4.4 Current policy relevance

In August 2026 DHS published a final rule rescinding the 2022 framework, effective September 18, 2026, which returns public charge adjudication to case-by-case discretion (Section 2.2). These results suggest

that a comparable episode would produce larger enrollment losses if prior fear has not fully dissipated. The asymmetry between scaring families away from programs and bringing them back implies that the cumulative effect of repeated restrictive episodes may exceed any single episode - a pattern the cumulative disadvantage literature would predict, since each episode compounds the effects of the last (DiPrete and Eirich, 2006). State-level protective policies may partially buffer federal chilling.

6.5 Limitations

The ACS mixed-status definition pools families with very different vulnerability levels, from those with undocumented parents to lawful permanent residents to temporary visa holders, which likely attenuates the estimate relative to the effect among the most vulnerable families. Additionally, as many other immigration enforcement actions were conducted during the study period, I am unable to discern if effects are from other enforcement shocks rather than public-charge exposure specifically. My comparison of children whose parent is a noncitizen with children whose parents are citizens but who live with a noncitizen adult relative cannot distinguish the two channels, and cannot detect a difference smaller than about 3.4 pp (Appendix E).

6.5.1 Pre-trends, timing, and state-environment sensitivity

As noted in previous sections, there is no well-dated intervention to assign pre-post timing to. A draft executive order leaked in January 2017, the proposal came in September 2018, and implementation and vacatur followed over three more years, with continuous enrollment overlapping most of the post period. Part of what I treat as the pre-period may already contain a response, and the phase pattern does not map cleanly onto the legal rule. The 2017 lead under the 2018 reference is +0.15 points (SE 0.53) and indistinguishable from zero (Appendix B), which is some evidence against a large response in the leak year. A partially exposed pre-period would bias the estimate toward zero rather than away from it.

6.5.2 Pre-trends and counterfactuals

Mixed-citizen families and all-citizen families may have been on distinctly different Medicaid coverage trajectories in the pre-public charge period. Rambachan-Roth breakdown value of 0.20 to 0.25 suggests the estimate does not survive a post-proposal differential trend much larger than a fifth of the largest pre-

period year-to-year movement. The non-expansion-state sample and two of the parent-pair contrasts reject their pre-trend tests, and the alternative comparison group collapses to zero without the 2020 file.

6.5.3 Post-period contamination

The post-announcement period includes several overlapping shocks: the public charge rule, COVID-19, the continuous enrollment provision, and the Medicaid unwinding. Difference-in-differences nets out common shocks affecting both groups equally, but any shock differentially affecting mixed-status families could bias the estimate, and the pandemic did affect immigrant communities disproportionately through infection rates, economic disruption, and access to services. Splitting the post period and re-estimating with alternative treatment timing provide partial leverage but cannot fully resolve the confounding.

6.5.4 Spillovers

Chilling is, by definition, a spillover effect - and as such are not confined to one group. California naturalized citizens, whom no public charge determination should affect, report a 13.6% avoidance share (Section 2.3), so comparison families nationally may themselves contain households with predilection for avoiding enrollment. If this is true, my differential design understates the total response.

6.5.5 External validity

As the 2026 rule even gave even further discretion to immigration enforcement officers to designate applicants as a public charge, it is difficult to apply these estimates in context of the new policy. While the rule should hypothetically reduce enrollment even further, this paper's estimates may be limited to providing an absolute lower bound of the effect of the August 2026 rule.

7. Conclusion

The public charge episode shows how a policy targeted at immigration status can reshape health coverage for populations well outside its formal legal scope. The mechanism is not legal exclusion - the populations most affected by the chilling effect were never subject to the rule - but rather, fear: fear

amplified by policy complexity, transmitted through immigrant networks, and made possible by an administrative environment that scrutinizes immigrant applicants more intensively than native-born citizens.

I estimate that low-income U.S. citizen children in mixed-status families experienced a 2.0 percentage point decline in Medicaid enrollment after the public charge rule's announcement, an effect that persisted through the rule's reversal and deepened during the Medicaid unwinding. California provides supporting evidence that chilling fears spilled over to naturalized citizens, a group with no direct legal vulnerability to the rule, and that immigrant groups face a markedly more burdensome documentation environment when seeking health coverage. For immigration policy, they show that the collateral effects of restrictive rules on health coverage are substantial, measurable, and persistent. For health policy, they show that enrollment infrastructure and administrative design can either amplify or attenuate the chilling effects of policy shocks. For both, they point to a basic asymmetry: it is easier to generate fear than to undo its effects.

As public charge rule has been recently reintroduced with a stronger version that could cause even more significant chilling effects, it should be evaluated as child health and Medicaid policy, not only as immigration policy.

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Appendix

Appendix A. Sample, exposure, and outcome definitions

Table A1 reconciles the two frames referenced in Section 3.1 and gives the exclusion counts at every step.

Table A1. ACS Children Sample flow, 2014-2024

Step	Child-years	Note
All children ages 0-18 in the parent-configuration frame	7,260,092	all-children frame before any restriction
U.S.-citizen child (CITIZEN in 0, 1, 2)	7,093,034	citizenship restriction
U.S.-born citizen child only (CITIZEN = 0)	6,941,159	alternative child rule
Citizen child with at least one identified resident parent	6,789,814	analysis universe
... with valid POVERTY	6,789,814	non-binding at this step
... below 200% FPL	2,194,701	
... pooled 2014-2022 window	1,841,358	primary estimation sample
U.S.-born branch, same chain, 2014-2022	1,804,558	definitional robustness sample

Notes: Both figures cited in Section 3.1 are child-years and not children. The gap between 7,260,092 and 6,789,814 is 470,278 child-years: 167,058 noncitizen children removed by the citizenship rule and 303,220 citizen children with no resident-parent pointer, or 4.27% of citizen children. Within the primary estimation sample, 1,804,558 child-years are born in the United States, 18,943 are born abroad to U.S. parents, and 17,857 are naturalized. Author's calculations from American Community Survey microdata (IPUMS USA, 2014-2024).

The outcome is the edited IPUMS HINSCAID, which indicates whether the person was covered at interview by Medicaid, Medical Assistance, or any other kind of government-assistance plan for those with low incomes or a disability. In the 2014-2022 primary sample the data-quality flag distribution is 88.55% not allocated, 0.00% coverage removed by logical edit, 3.35% coverage added by logical edit, and 8.11% allocated. Codes indicating an edit are retained and only the allocation code is treated as an imputed response. Private coverage is HCOVPRIV, which counts TRICARE and other military coverage; any public coverage is HCOVPUB, which counts Medicare and veterans' coverage; uninsurance is the complement of HCOVANY.

Appendix B. Event-study coefficients, phases, and pre-trend tests

Table B1. ACS Children Event-study coefficients, Medicaid/CHIP coverage, below 200% FPL

Survey year	2018 reference	2017 reference	2017 reference, state-by-year FE
2014	+0.85 (0.89)	+0.71 (0.64)	+1.38** (0.56)
2015	+1.41 (0.87)	+1.27* (0.65)	+1.10 (0.67)
2016	+0.40 (1.02)	+0.25 (0.80)	+0.29 (0.84)
2017	+0.15 (0.53)	reference	reference
2018	reference	-0.15 (0.53)	+0.03 (0.50)
2019	-0.92 (0.58)	-1.06 (0.70)	-0.79 (0.64)
2020	-2.47*** (0.89)	-2.62** (1.02)	-2.91*** (0.86)
2021	-1.57** (0.63)	-1.71*** (0.58)	-1.55*** (0.57)
2022	-0.80 (0.61)	-0.95* (0.48)	-0.55 (0.52)
2023	-1.82*** (0.55)	-1.96*** (0.55)	-1.79*** (0.56)
2024	-3.00*** (0.88)	-3.14*** (1.11)	-3.00*** (0.66)

Notes: Standard errors clustered on state in parentheses. Estimated over 2014-2024 with the Section 3.5 controls. Joint Wald tests on the leads strictly before the reference year: $p = 0.175$ under the 2018 reference (four leads), $p = 0.103$ under the 2017 reference (three leads), and $p = 0.089$ under state-by-year fixed effects. Every joint test in this table is a property of the window it covers; Appendix C reports the same test on a pre-period beginning in 2008. The 2020 column entries come from the experimental ACS file, which the baseline pooled estimate includes and the excluding-2020 specification drops; the 2023-2024 entries fall in the unwinding period and are excluded from every pooled estimate. Dropping 2020 from the event study leaves the remaining coefficients unchanged to two decimal places. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Author's calculations from American Community Survey microdata (IPUMS USA, 2014-2024).

Table B2. ACS Children Pooled and policy-specific estimates, Medicaid/CHIP coverage

Specification	Estimate (pp)	SE	p	N
Pooled 2014-2022, all years	-2.00	0.56	<0.001	1,841,358
Pooled, dropping 2018 entirely	-2.14	0.67	0.003	1,632,458
Pooled, dropping 2020	-1.67	0.44	<0.001	1,697,491
Pooled, dropping 2018 and 2020	-1.81	0.55	0.002	1,488,591
Pooled 2016-2022	-1.63	0.48	0.001	1,361,769
Phase: 2018 transition	-0.72	0.76	0.345	1,841,358
Phase: 2019 proposal to finalization	-1.63	0.91	0.082	1,841,358
Phase: 2020 implementation (experimental ACS)	-3.17	1.25	0.014	1,841,358
Phase: 2021-2022 post-vacatur and continuous	-1.91	0.45	<0.001	1,841,358

enrollment				
Descriptive split: 2019-2022	-2.01	0.56	<0.001	2,194,701
Descriptive split: 2023-2024 unwinding	-2.98	0.86	0.001	2,194,701

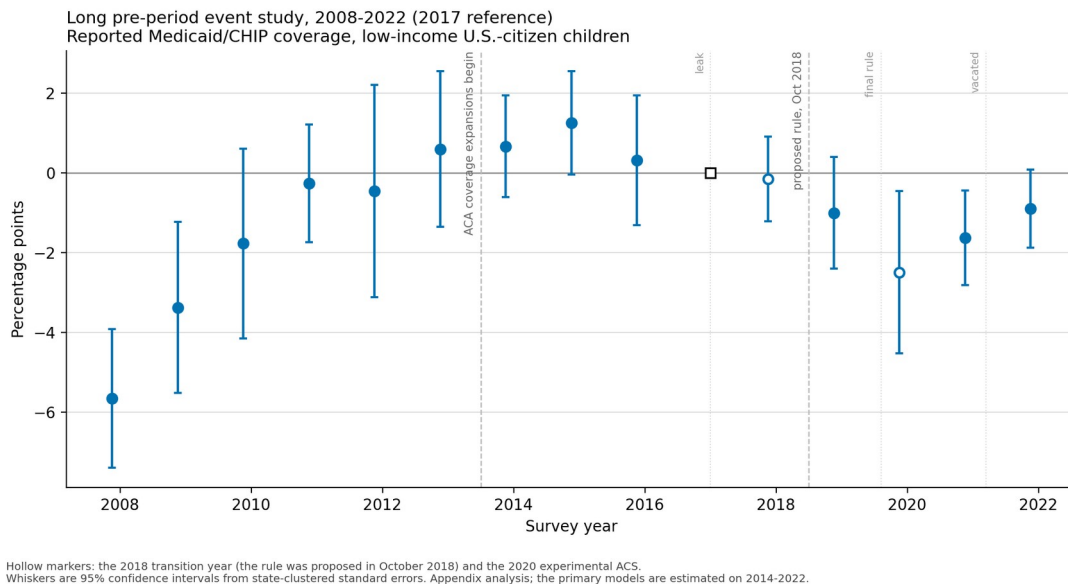
Notes: Phase rows come from one regression with 2014-2017 as the pre-period and separate indicators for 2018, 2019, 2020, and 2021-2022. A joint test that the three post phases are equal gives $p = 0.094$; dropping both 2018 and the experimental 2020 file leaves 2019 at -1.62 and 2021-2022 at -1.91 with an equality test of $p = 0.700$. Secondary outcomes by phase (2018 / 2019 / 2020 / 2021-2022) are +0.30, +0.61, +0.91, +0.29 for uninsurance and +0.35, +1.04, +2.45, +1.71 for private coverage. Author's calculations from American Community Survey microdata (IPUMS USA, 2014-2024).

Appendix C. The long pre-period, 2008-2022

This appendix rebuilds the analysis frame from a single IPUMS extract covering ACS one-year samples from 2008 forward, on the native person-weight scale, with every definition held identical to the primary build. Group-quarters records are excluded, children are ages 0 to 18, the child rule is CITIZEN in 0, 1, 2, resident parents come from the household-roster pointers, and the outcome is the edited HINSCAID. Public use microdata areas are not used anywhere in the module because their boundaries change in 2012 and 2022. The build runs from 10,215,033 children to 9,549,815 citizen children with a resident parent and valid poverty information, of whom 3,335,921 fall below 200% FPL, including 1,494,563 in the added 2008-2013 block.

Two checks were run before any estimation. Every mother pointer resolves to a person coded female and every father pointer to a person coded male, across 9,324,236 and 7,833,664 pointers with no unmatched targets. In addition, the rebuilt 2014-2022 sample contains 1,841,358 child-years, identical to the primary estimation sample, and returns -2.0033 points with a standard error of 0.5596. The primary estimate is therefore reproduced from an independently constructed source file. The U.S.-born-only variant reproduces at -1.912 points on 1,804,558 child-years.

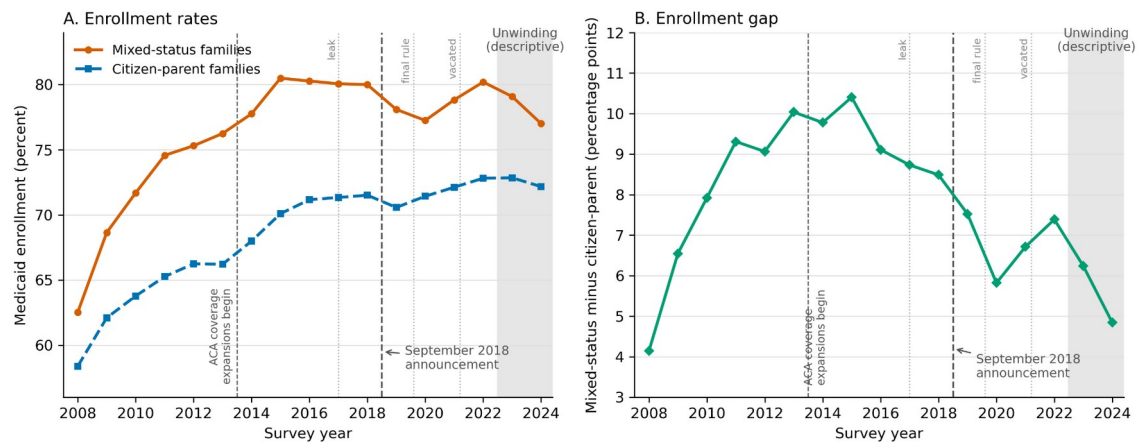
Figure C1. Event study on the long pre-period, 2008-2022



Notes: Event-study coefficients with 2017 as the omitted reference and 2018 carrying its own transition coefficient, estimated on the rebuilt 2008-2022 frame with state and year fixed effects, the Section 3.5 controls, and standard errors clustered on state. Bars are 95% confidence intervals. Dashed lines mark the September 2018 proposal and the start of the ACA coverage expansions, and lighter dotted lines mark the January 2017 draft-order leak, the August 2019 final rule, and the March 2021 vacatur (Table 1). Author's calculations from American Community Survey microdata (IPUMS USA, 2008-2022).

Figure C2 plots the raw series behind this event study. The gap between the two groups rises from 4.1 points in 2008 to a 10.4-point peak in 2015 and drifts down over the three years before the proposal. That is the movement Figure C1 formalizes: sharp divergence over the coverage-expansion years, a flat 2014–2017 stretch, and a post-2018 departure from that stretch.

Figure C2. Raw Medicaid/CHIP coverage trends on the extended pre-period, 2008–2024



Notes: Panel A plots survey-weighted Medicaid/CHIP coverage rates for the two groups of U.S.-citizen children below 200% FPL; note the truncated vertical axis. Panel B plots the difference between the series in percentage points. Rates for 2008–2022 are computed from the single 2008–2022 extract underlying Figure C1, and rates for 2023–2024 are taken from the primary analysis file; on the overlapping years the two sources agree exactly for the mixed-status series and within 0.35 percentage points for the citizen-parent series. The dashed lines mark the September 2018 proposal and the start of the ACA coverage expansions, the lighter dotted lines mark the January 2017 draft-order leak, the August 2019 final rule, and the March 2021 vacatur (Table 1), and the shaded region covers the 2023–2024 unwinding years. Author’s calculations from American Community Survey microdata (IPUMS USA, 2008–2024).

Table C1. ACS Children Long pre-period event study and placebo break dates

Panel A. Event-study coefficients, 2017 reference

Survey year	Estimate (pp)	SE	p
2008	-5.65	0.86	<0.001
2009	-3.38	1.07	0.003
2010	-1.77	1.19	0.141
2011	-0.26	0.74	0.723
2012	-0.46	1.33	0.731
2013	+0.59	0.97	0.546
2014	+0.66	0.64	0.303
2015	+1.26	0.65	0.058
2016	+0.31	0.81	0.703
2018 (transition)	-0.15	0.53	0.774
2019	-1.01	0.70	0.155
2020 (experimental ACS)	-2.50	1.01	0.017
2021	-1.63	0.59	0.008
2022	-0.90	0.49	0.072

Panel B. Placebo break dates estimated on pre-period data only

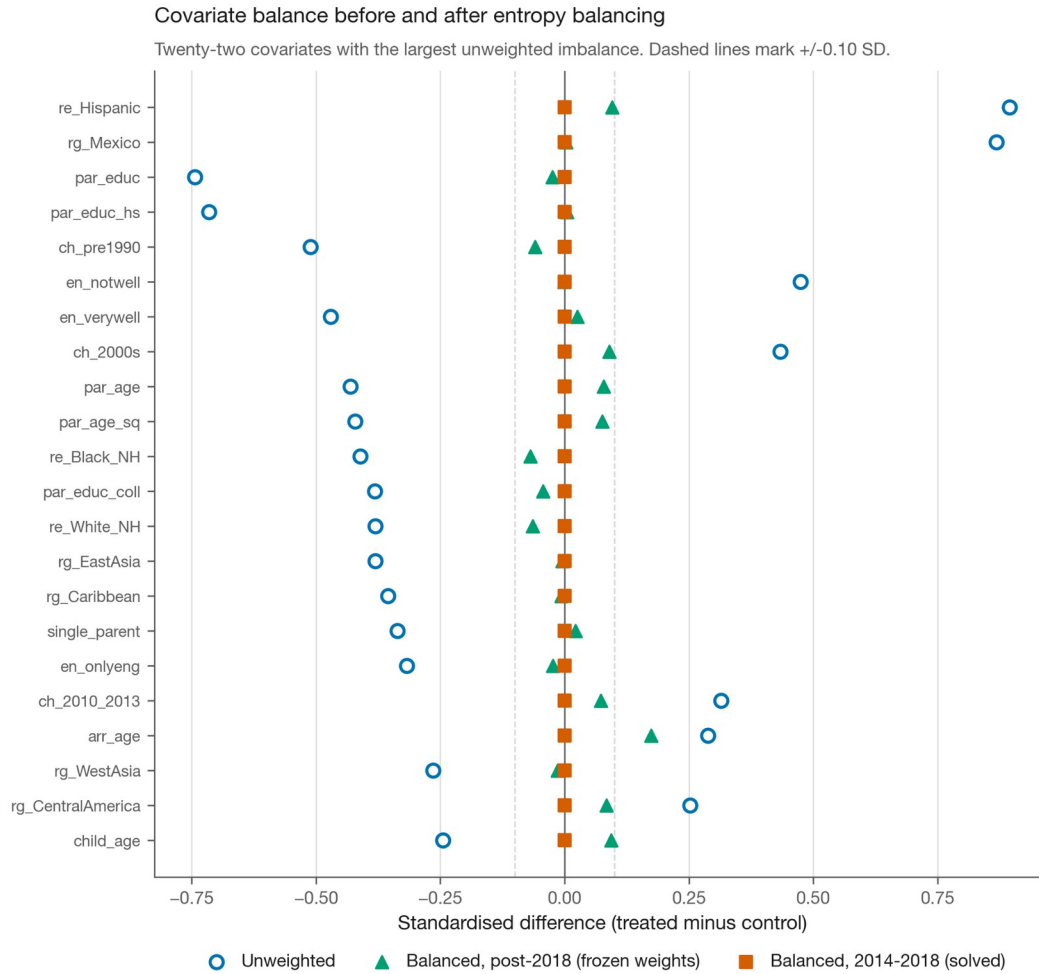
Fake break	Window	Medicaid/CHIP	SE	p	Uninsured	p
Post = 2012	2008-2018	+3.03	0.47	<0.001	-3.39	<0.001
Post = 2014	2008-2018	+2.19	0.68	0.002	-2.74	<0.001
Post = 2016	2008-2018	+1.14	0.67	0.094	-2.21	<0.001
Post = 2011	2008-2013	+3.54	0.53	<0.001	-	-
Post = 2016	2014-2018	-0.90	0.48	0.069	-0.31	0.148
Post = 2016	2014-2017	-0.81	0.40	0.052	-0.49	0.016

Notes: Joint Wald tests on the leads: $p < 0.001$ for all leads 2008-2016 (nine leads), $p < 0.001$ for the added 2008-2013 leads alone (six leads), and $p = 0.1195$ for the 2014-2016 leads alone (three leads). Differential linear pre-trends estimated on pre-period data are +1.19 points per year over 2008-2013, +0.59 over 2008-2017, and +0.47 over 2008-2018. The raw weighted gap in Medicaid/CHIP coverage between the two groups runs 4.14 (2008), 6.55, 7.92, 9.31, 9.06, 10.04 (2013), 9.78, 10.40, 9.11, 8.73 (2017), 8.49, 7.53, 5.82, 6.71, 7.39 (2022). Pooled estimates over the long window are -0.76 (SE 0.86, $p = 0.383$) for 2008-2022 and -1.54 (SE 0.85, $p = 0.075$) for 2010-2022. Dashes denote cells not estimated. Both are attenuated mechanically, because the post period is differenced against a pre-period mean that includes the strongly non-parallel 2008-2010 block. Neither is evidence that the effect is smaller than the 2014-2022 estimate. The private-coverage placebo at a fake 2016 break on the 2014-2018 window is +1.22 ($p = 0.009$). Author's calculations from American Community Survey microdata (IPUMS USA, 2008-2022).

Power of the 2014-2017 lead test. Following Roth (2022), I compute the power of the joint test on the 2014-2017 leads against linear violations of parallel trends, using the event-study coefficients and their estimated covariance matrix. The linear differential trend that would generate the full baseline estimate of -2.00 points has a slope of -0.80 points per year, and the lead test would reject it with 95.8% power. The slopes the test can detect at 50% and 80% power are 0.41 and 0.60 points per year. The flat 2014-2017 stretch is therefore informative against trends of the size the estimate implies, and it places little restriction on trends much smaller than that.

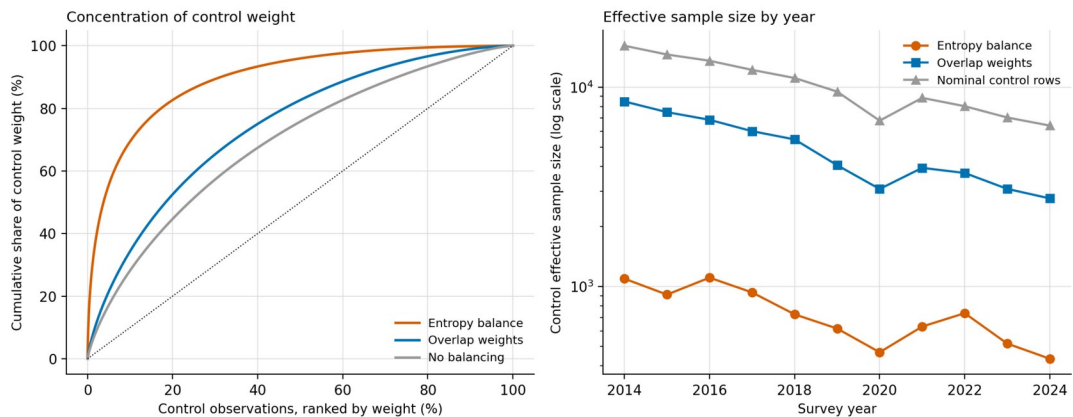
Appendix D. Long-naturalized comparison: balance, weights, and overlap

Figure D1. Covariate balance, long-naturalized-parent comparison



Notes: Standardized differences between children of noncitizen parents and children of long-naturalized parents, before and after entropy balancing on 89 covariate constraints covering parents' origin region, arrival cohort, age at arrival, education, English proficiency, child age, family structure, state, and income position. Balancing weights are solved on 2014-2018 and frozen thereafter. Author's calculations from American Community Survey microdata (IPUMS USA, 2014-2024).

Figure D2. Weight diagnostics, entropy balancing and overlap weighting



Notes: Distribution of comparison-group weights under the entropy-balanced and overlap-weighted designs. Author's calculations from American Community Survey microdata (IPUMS USA, 2014-2022).

Table D1. Long-naturalized design, weighting schemes and diagnostics

Quantity or variant	Entropy balancing	Overlap weights
Comparison-group effective sample size	7,774 of 114,080 (6.8%)	54,373 (47.7%)
Maximum weight divided by mean weight	202.9	19.8
99th-percentile weight	6,032.8	319.1
Share of comparison weight in the top 1%	28.1%	6.6%
Share of comparison weight in the top 5%	55.6%	21.5%
Maximum absolute standardized difference, pooled	0.089	0.080
Maximum absolute standardized difference, post-2018 under frozen weights	0.260	0.178

Notes: Comparison-group effective sample size by year runs 435 to 1,108 under entropy balancing. By state the median is 69, eleven states have an effective sample size below 30, thirty-two below 100, and California alone carries 26.1% of the balanced comparison weight, with Texas at 17.8%. The balanced design therefore compares roughly 300,000 children with a noncitizen parent against the statistical equivalent of about 7,800 comparison children concentrated in two states. Secondary outcomes are -1.71 (balanced), -0.84 (overlap), and -1.46 (common support, $p = 0.022$) for any public coverage, and -1.62, -0.48, and +0.27 for uninsurance, none of the uninsurance estimates statistically distinguishable from zero. Excluding the experimental 2020 ACS gives +0.02 (balanced), +0.01 (overlap), -0.73 (common support), and -2.08 (unbalanced, $p = 0.005$). Balanced event-study coefficients under the 2018 reference are +0.52 (2019), -6.55 (2020), -1.25 (2021), and +0.62 (2022); joint pre-trend tests give $p = 0.42, 0.33$, and 0.70 across the three weighted versions. A geographic-block sensitivity moves the balanced estimate from -2.20 with no block to -1.57 with full state dummies, with standard errors near 1.7 points throughout, and trimming the weights at the 95th percentile gives -2.40 ($p = 0.009$) while breaking exact balance at 0.22 standard deviations. Author's calculations from American Community Survey microdata (IPUMS USA, 2014-2022).

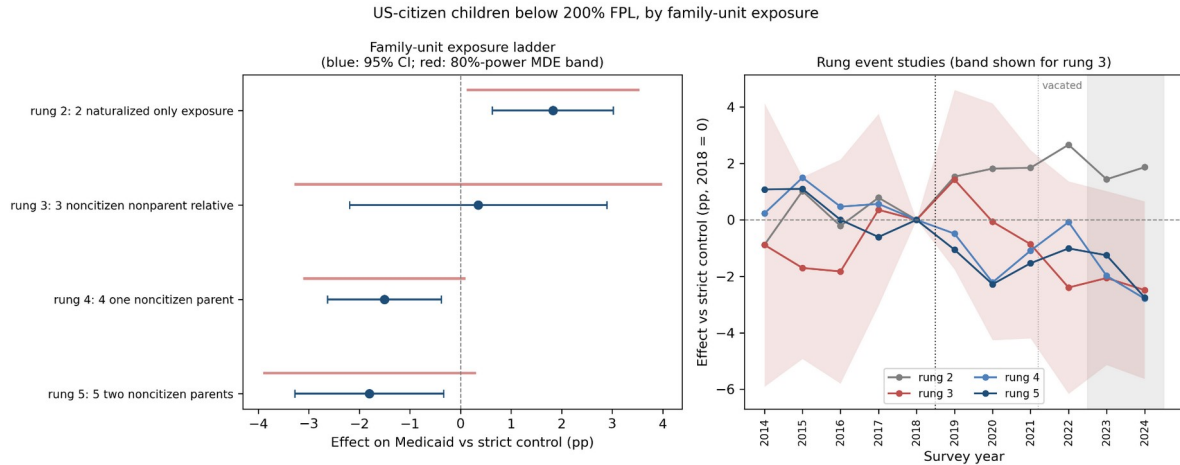
Table D2. ACS Children Change in Medicaid/CHIP coverage over time, parental citizenship groups

Specification	Group or contrast	Estimate (pp)	SE	p	N
Four-level, reference U.S.-born-parent	Long-naturalized parent (by 2013)	+0.99	0.74	0.190	1,832,638
Four-level, reference U.S.-born-parent	Recently naturalized parent (2014-2018)	+0.11	1.14	0.921	1,832,638
Four-level, reference U.S.-born-parent	Noncitizen parent	-1.83	0.60	0.004	1,832,638
Four-level, pairwise contrast	Noncitizen vs long-naturalized	-2.82	0.63	<0.001	1,832,638
Pooled naturalized comparison	Any naturalized parent, as classified each year	+1.91	0.69	0.008	1,841,358

Notes: The four-level rows come from one multi-group regression on the below-200% FPL sample with U.S.-born-parent families as the reference; families whose parent naturalized in 2019 or later are excluded because their group assignment would be determined after the proposal. The pooled row classifies a family as naturalized in each year on that year's reported status, which is why it returns a positive coefficient: the group's composition drifts toward recent naturalizers with higher baseline coverage, and splitting on naturalization timing removes the anomaly. Weighted least squares, state and year fixed effects, standard errors clustered on state, 2014-2022. Author's calculations from American Community Survey microdata (IPUMS USA, 2014-2022).

Appendix E. Exposure gradient: family-unit ladder and multiplicity adjustment

Figure E1. Family-unit exposure ladder, Medicaid/CHIP coverage



Notes: Pooled estimates for each rung of the family-unit exposure ladder relative to a comparison group in which every member of the child's family unit is a U.S.-born citizen, with 95% confidence intervals. Pooled window 2014-2022, below 200% FPL, weighted least squares with state and year fixed effects and standard errors clustered on state. In the event-study panel, the dotted line marks the September 2018 proposal and the lighter dotted line marks the March 2021 vacatur. Author's calculations from American Community Survey microdata (IPUMS USA, 2014-2022).

Table E1. ACS Children Family-unit exposure ladder and comparison-group definitions

Rung or comparison-group definition	Estimate (pp)	SE	p	MDE (80%)
Naturalized adult in family unit, no noncitizen member	+1.83	0.61	0.004	-
Noncitizen non-parent relative in family unit	+0.35	1.30	0.786	3.64
One noncitizen parent	-1.51	0.57	0.012	-
Two noncitizen parents	-1.80	0.75	0.020	-
Contrast: one noncitizen parent vs noncitizen non-parent relative	-1.86	1.13	0.101	3.18
Baseline contrast, comparison group = every citizen child with no noncitizen parent	-2.00	0.56	<0.001	-
Baseline contrast, comparison group = U.S.-born-parent families only	-1.79	0.60	0.004	-
Baseline contrast, comparison group = every family-unit member a U.S.-born citizen	-1.81	0.61	0.005	-

Notes: The ladder is estimated on 1,806,940 children with the strict comparison group as reference, which requires every member of the child's family unit to be a U.S.-born citizen; the ladder uses family-unit relationships instead of household membership, so an unrelated household member cannot classify a child as exposed. All five rungs pass their joint pre-trend tests ($p = 0.10$ to 0.37). The non-parent-relative rung would test whether household network exposure matters separately from the exposure of the adult who files the application. However, it is underpowered for that purpose: its minimum detectable effect of 3.64 points exceeds the prior-literature magnitude it is being asked to detect. The MDE column reports 80%-power minimum detectable effects for the null estimates and is blank where the estimate is distinguishable from zero. Under state-by-year fixed effects the four rungs are +1.82, +0.44, -1.43, and -1.74. Author's calculations from American Community Survey microdata (IPUMS USA, 2014-2022).

Table E2. Multiplicity-adjusted inference for the parent-pair contrast family

Contrast	Estimate (pp)	p raw	p bootstrap	p Holm	p Romano-Wolf	p Bonferroni	Pre-trend
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NN vs ZZ	-4.31	0.00004	<0.0001	0.00025	0.0032	0.00025	rejects (0.015)
single N vs single Z	-3.23	0.00039	<0.0001	0.00194	0.0154	0.00233	passes (0.254)
Linear dose	-1.30	0.00204	0.0398	0.00817	0.0432	0.01225	-
ZN vs ZZ	-2.63	0.00420	0.0032	0.01261	0.0432	0.02521	passes (0.585)
BN vs BZ	-3.65	0.00990	0.0126	0.01979	0.0432	0.05938	passes (0.436)
NN vs BN	-0.86	0.35776	0.4032	0.35776	0.4032	1.00000	rejects (0.0002)
Prespecified confirmatory family of three							
single N vs single Z	-3.23	0.00039	<0.0001	0.00116	0.0001	0.00116	passes
ZN vs ZZ	-2.63	0.00420	0.0032	0.00840	0.0073	0.01261	passes
BN vs BZ	-3.65	0.00990	0.0126	0.00990	0.0126	0.02969	passes

Notes: Standard errors are 1.36 (BN vs BZ), 0.88 (ZN vs ZZ), 0.85 (single N vs single Z), 0.96 (NN vs ZZ), 0.92 (NN vs BN), and 0.40 (linear dose), all clustered on state. The six contrasts come from three separate regressions, so inference uses a joint wild cluster bootstrap with Rademacher weights, clustered on the 51 states, unrestricted and recentred, with 9,999 replications and the same cluster draws applied to all three models, followed by Romano-Wolf stepdown and Holm corrections on the studentized statistics. Bootstrap means equal the point estimates and bootstrap standard deviations sit within 2% of the analytical clustered standard errors. Five of the six contrasts survive Romano-Wolf and Holm at 5%; the exception, NN vs BN, also fails its pre-trend test. Bonferroni alone would drop BN vs BZ, which is why the stepdown procedures are reported. The linear dose is the contrast most sensitive to the choice of inference, moving from $p = 0.002$ analytically to $p = 0.040$ under the bootstrap, and is read as a summary of direction rather than as a dose-response estimate. Author's calculations from American Community Survey microdata (IPUMS USA, 2014-2022).

Table E3. ACS Children Confirmatory contrasts under added year interactions, all-configuration parameterization

Added fixed effects	BN vs BZ	ZN vs ZZ	Single N vs single Z
Base: state and year FE	-3.65*** (1.36)	-2.63*** (0.88)	-3.23*** (0.85)
Parent origin region-by-year	-3.25** (1.46)	-1.86** (0.88)	-2.83*** (0.82)
Parent arrival cohort-by-year	-3.39** (1.33)	-2.80*** (0.89)	-2.99*** (0.83)
Parent English proficiency-by-year	-3.37** (1.39)	-2.58*** (0.83)	-2.59*** (0.88)
Origin- and arrival-cohort-by-year	-2.87* (1.44)	-1.96** (0.89)	-2.30*** (0.80)
Origin-, arrival-cohort- and English-proficiency-by-year	-2.76* (1.47)	-1.96** (0.88)	-2.02** (0.84)

Notes: Standard errors clustered on state in parentheses. These rows keep the Table 8 parameterization, in which all six two-parent configurations (or all three single-parent configurations) enter one regression with the U.S.-born-parent group omitted and the contrast recovered as a linear combination. Table 9 reports the same variants estimated on the two contrasted configurations alone. The two parameterizations differ by at most 0.12 points at the base specification. The reference cell in this parameterization is nearly spanned by the origin and arrival-cohort fixed effects, since families with no foreign-born parent are by construction the omitted group: 312 non-reference rows sit in the two-parent reference cell and 183 in the single-parent one, so the reference group's time path is held in place by a small number of observations, which is why Table 9 uses the pair-only form. Recoding the reference into its own origin and cohort cell and treating the resulting dropped column as the normalization it is

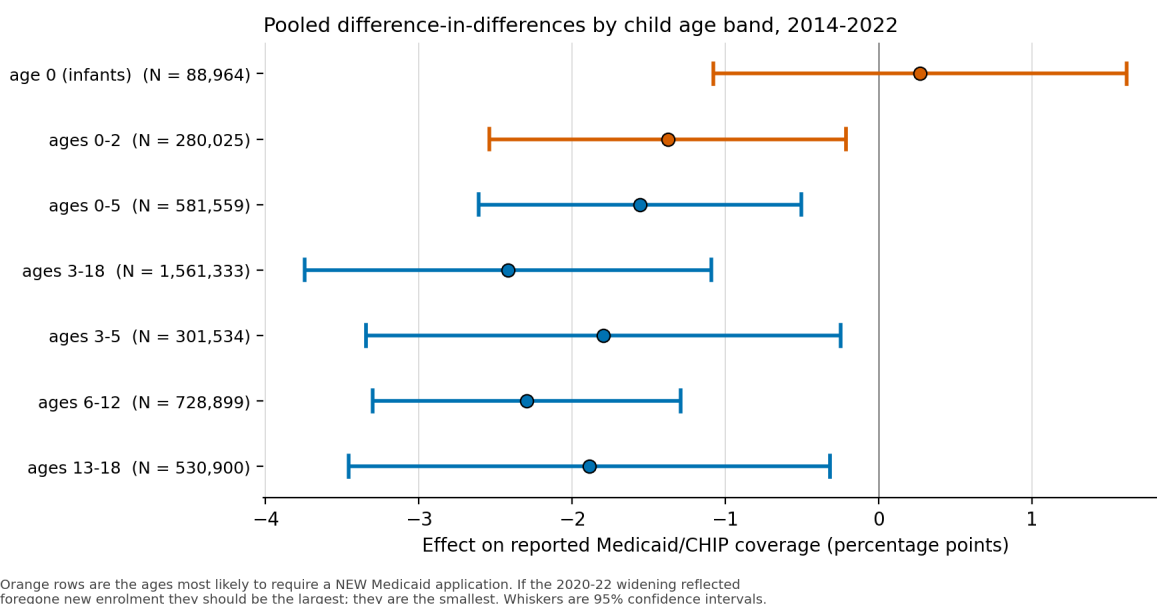
moves the contrasts by at most 0.13 points. Below 200% FPL, pooled 2014-2022. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Author's calculations from American Community Survey microdata (IPUMS USA, 2014-2022).

Appendix F. Continuous enrollment, age bands, and baseline headroom

This appendix examines the 2020-2022 window, in which states receiving enhanced federal matching funds could not disenroll most Medicaid beneficiaries. With routine disenrollment constrained, a gap opening in that window would have to come substantially from families not enrolling children who would otherwise have been enrolled, and children who need a new application are concentrated at the youngest ages. The age-band estimates run against that account: the estimate is +0.27 points for infants (SE 0.67, $p = 0.69$) against -2.42 for ages 3 to 18 (SE 0.66, $p < 0.001$), the extra effect in the younger band is +2.64 points at the age-0 split ($p = 0.018$) and +0.88 at the age-5 split ($p = 0.050$), and restricting to the continuous-enrollment window itself gives +2.80 ($p = 0.033$).

An alternative account fits the same window. Citizen-parent children started with 6 to 9 points more room to gain coverage in every age band, and under a policy that protects existing enrollees the group with more people already covered gains less; I refer to this room as baseline headroom. Across 89 state-by-group cells the raw differential change from 2018 to 2021-2022 is -1.43 points (SE 0.58, $p = 0.017$), and adding a control for each cell's 2018 baseline level takes it to -0.16 points (SE 0.64, $p = 0.809$). Baseline coverage accounts descriptively for most of that differential, though it also proxies other group-by-state differences. I therefore do not attribute the 2020-2022 movement to foregone new enrollment.

Figure F1. Estimates by child age band



Notes: Pooled estimates by child age band with 95% confidence intervals, below 200% FPL, 2014-2022. Author's calculations from American Community Survey microdata (IPUMS USA, 2014-2022).

Table F1. ACS Children Age bands, triple differences, and baseline headroom

Panel A. Pooled estimates by age band

Age band	Estimate (pp)	SE	p	N	Excluding 2020	2018 rate, mixed-status	2018 rate, citizen-parent
Age 0	+0.27	0.67	0.687	88,964	+0.66	82.92	76.80

Ages 0-2	-1.38	0.58	0.022	280,025	-1.21	82.67	75.23
Ages 0-5	-1.56	0.52	0.005	581,559	-1.38	82.79	74.58
Ages 3-5	-1.80	0.77	0.024	301,534	-1.62	82.93	73.97
Ages 3-18	-2.42	0.66	<0.001	1,561,333	-2.05	79.42	70.79
Ages 6-12	-2.29	0.50	<0.001	728,899	-2.02	80.75	71.91
Ages 13-18	-1.89	0.78	0.020	530,900	-1.31	75.00	67.26

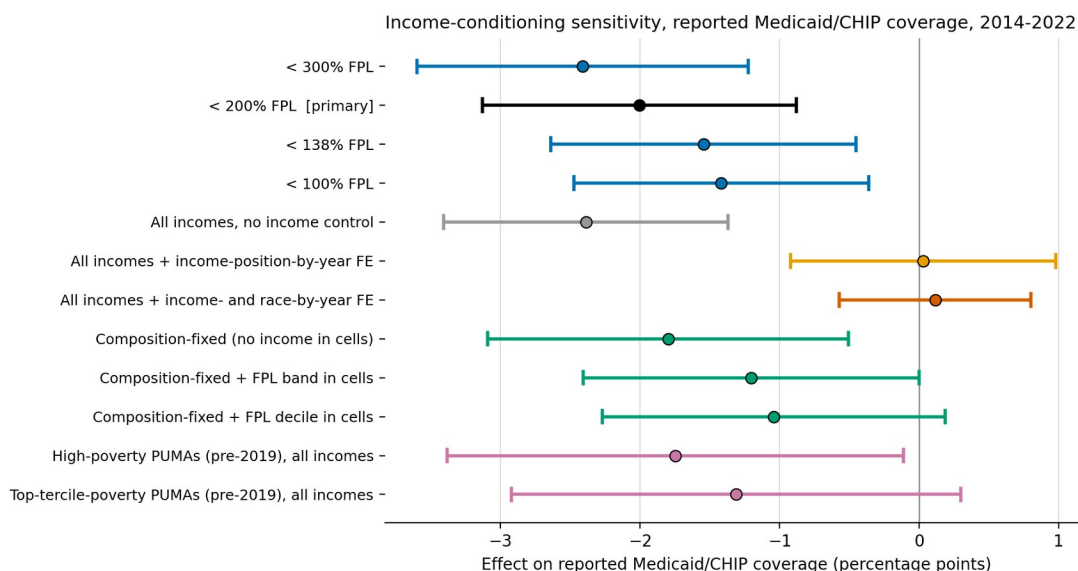
Panel B. Triple differences and the headroom check

Test	Estimate (pp)	SE	p
Extra effect in the younger band, ages 0-2 vs 3-18	+0.99	0.78	0.213
Extra effect in the younger band, age 0 vs 1-18	+2.64	1.08	0.018
Extra effect in the younger band, ages 0-5 vs 6-18	+0.88	0.44	0.050
Age 0 vs 1-18, continuous-enrollment window only	+2.80	1.28	0.033
State-by-group differential change, 2018 to 2021-2022	-1.43	0.58	0.017
Same, controlling for the 2018 baseline level	-0.16	0.64	0.809

Notes: A negative triple difference would support foregone new enrollment as the channel operating during continuous enrollment; every estimate in Panel B has the opposite sign, and the age-0 split is significant. The continuous-enrollment window rows compare 2021-2022 with 2014-2018, dropping 2019 and the experimental 2020 file. The headroom rows are estimated on 89 state-by-group cells, weighted and clustered on state; the coefficient on the 2018 baseline level is -0.150 ($p = 0.004$), so cells that started higher grew less. A third specification adding state fixed effects is collinear by construction, because with state fixed effects the only remaining variation in the baseline level is the within-state gap between the two groups, and it is not reported as evidence. Phase coefficients by band confirm the pattern: infants show +0.68 (2019), -1.16 (2020), and +0.40 (2021-2022), none significant, while ages 3-18 show -1.99, -3.74, and -2.34, all significant. Author's calculations from American Community Survey microdata (IPUMS USA, 2014-2022).

Appendix G. Income conditioning

Figure G1. Estimates under alternative income definitions



Notes: Pooled estimates under the income-band, all-income, composition-fixed, and geography-based sample definitions, with 95% confidence intervals. Author's calculations from American Community Survey microdata (IPUMS USA, 2014-2022).

Table G1. ACS Children Income conditioning and sample-selection diagnostics

Specification	Estimate (pp)	SE	p	N
Selection diagnostic: movement in the sample-membership rule				
Outcome = below 200% FPL	-4.33	0.31	<0.001	5,550,696
Outcome = below 138% FPL	-4.90	0.37	<0.001	5,550,696
Outcome = below 300% FPL	-1.57	0.26	<0.001	5,550,696
Outcome = income-to-poverty ratio (FPL points)	+8.30	0.79	<0.001	5,550,696
All-income models				
No income control	-2.39	0.51	<0.001	5,550,696
Below-200% indicator as a static control	-0.22	0.53	0.675	5,550,696
Income-position-by-year FE, coarse bins	-0.02	0.49	0.970	5,550,696
Income-position-by-year FE, fine bins	+0.03	0.47	0.954	5,550,696
Income- and race-by-year FE	+0.11	0.34	0.740	5,550,696
Income-position-by-year FE, below 300% FPL	-0.64	0.60	0.297	2,804,622
Composition fixed on pre-period cells				
Origin, arrival cohort, parent education, child age, family structure	-1.80	0.64	0.007	1,841,358
Same, adding FPL band	-1.20	0.60	0.050	1,841,358
Same, adding FPL decile	-1.04	0.61	0.095	1,841,358

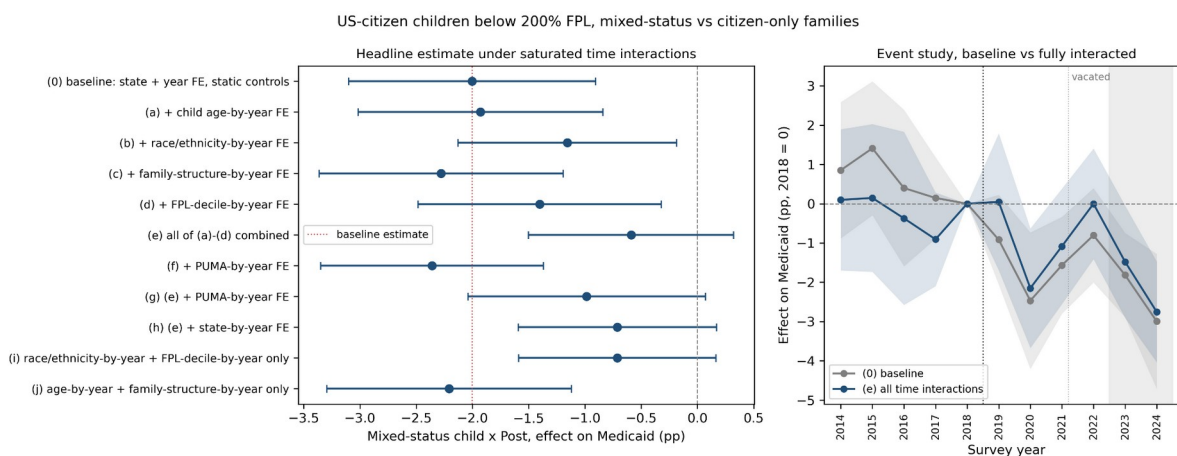
Geography-based low-income definition, no contemporaneous income screen

PUMAs above the median pre-2019 poverty share, all incomes, 2014-2021	-2.22	0.76	0.006	2,421,202
Same, adding income-position-by-year FE	-0.08	0.77	0.921	2,421,202
Same, adding race and ethnicity-by-year FE	-1.41	0.64	0.033	2,421,202
Same, adding state-by-year FE	-2.18	0.66	0.002	2,421,202
Same, reimposing the below-200% FPL screen	-1.79	0.77	0.025	1,127,069
Top-tercile PUMAs, all incomes, 2014-2021	-2.02	0.73	0.008	1,569,474
Top-quartile PUMAs, all incomes, 2014-2021	-1.78	0.65	0.009	1,141,134
Top-decile PUMAs, all incomes, 2014-2021	-1.49	0.90	0.108	446,150
States above the median poverty share, all incomes	-2.41	0.62	<0.001	3,692,991
Top-tercile states, all incomes	-3.26	0.76	<0.001	1,974,233
Income-screen indicator inside the high-poverty PUMA sample	-3.72	0.39	<0.001	2,421,202

Notes: The selection rows use the sample-membership indicator itself as the outcome; a nonzero coefficient means the primary sample's inclusion rule moves with the policy period. The income-to-poverty ratio row is in FPL points instead of percentage points. POVERTY is top-coded at 501 and 25.15% of the all-income sample sits at the top code, so exact deciles collapse: the coarse grid yields 8 distinct bins and the fine grid 15, and both are reported. The composition-fixed rows reweight mixed-status observations within each year to the pooled 2014-2018 mixed-status distribution over the stated cells, using 1,071, 2,880, and 7,816 cells respectively; excluding 2020 gives -1.54, -0.91, and -0.79. In the geography block each PUMA or state is classified once from pooled 2014-2018 data by its weighted share of citizen children below 200% FPL, and every child living there enters at any income. PUMA-classified rows cover 2014-2021 because PUMA boundaries were redrawn for the 2020 census and applied in the 2022 ACS, and only 1,442 of the 2,462 PUMA codes present in 2022 also appear in 2014-2018, so a pooled 2014-2022 PUMA classification would assign about a ninth of the sample the child-poverty class of a different area. State-classified rows keep the full 2014-2022 window because state codes are stable, and they rest on 26 clusters. The final row shows that the income response itself persists inside the fixed-geography sample: geography removes income from the membership rule without removing the underlying movement. Author's calculations from American Community Survey microdata (IPUMS USA, 2014-2022).

Appendix H. Richer time-interaction specifications

Figure H1. Estimates under characteristic-by-year fixed effects



Notes: Pooled estimates under progressively richer sets of characteristic-by-year fixed effects, with 95% confidence intervals. Below 200% FPL, 2014-2022, standard errors clustered on state. In the event-study panel, the dotted line marks the September 2018 proposal and the lighter dotted line marks the March 2021 vacatur. Author's calculations from American Community Survey microdata (IPUMS USA, 2014-2022).

Table H1. ACS Children Characteristic-by-year fixed effects

Specification	Estimate (pp)	SE	p	Share of treatment variation retained
Baseline: state and year FE, static controls	-2.00	0.56	<0.001	0.53
Child age-by-year FE	-1.93	0.56	0.001	0.53
Race and ethnicity-by-year FE	-1.16	0.50	0.024	0.39
Family-structure-by-year FE	-2.28	0.55	<0.001	0.46
FPL-decile-by-year FE	-1.40	0.55	0.014	0.53
PUMA-by-year FE	-2.36	0.51	<0.001	0.39
All four characteristic-by-year FE combined	-0.59	0.47	0.211	0.34
Combined plus PUMA-by-year FE	-0.98	0.54	0.073	0.31
Combined plus state-by-year FE	-0.71	0.45	0.120	0.34
Race and ethnicity-by-year plus FPL-decile-by-year only	-0.71	0.45	0.118	0.39
Child age-by-year plus family-structure-by-year only	-2.21	0.55	<0.001	0.46

Notes: Each row replaces the corresponding static control with its full interaction with survey year. Family structure is resident parents by own children by other family-unit adults; FPL deciles are computed within the below-200% FPL sample. The share of treatment variation retained is the share of the weighted variance of the mixed-status by post term that survives the fixed effects, so the attenuation is not a mechanical consequence of exhausting the variation. Under the combined specification the pre-proposal leads flatten from +0.85, +1.41, +0.40, and +0.15 to +0.10, +0.15, -0.37, and -0.91, and the post path is +0.05, -2.15, -1.08, and -0.00. Mixed-status children are 81% Hispanic against 22% of comparison children, which is why race and ethnicity-by-year effects absorb so much of the contrast. Author's calculations from American Community Survey microdata (IPUMS USA, 2014-2022).

Appendix I. Imputed likely undocumented parental status

The ACS does not observe immigration status, so the noncitizen-parent group pools lawful permanent residents, refugees and asylees, temporary visa holders, and undocumented parents. This appendix splits that group using a multiply imputed status classification and reports the resulting gradient, together with a diagnostic on whether the classification itself moves across the policy break. Every group label below is “imputed likely undocumented” or “imputed likely lawfully present”; nothing here is an observed legal-status classification.

Method. The standard residual approach classifies a foreign-born respondent as lawfully present partly because they receive a public benefit, and the benefit list in published implementations includes Medicaid. Using that algorithm in a study whose outcome is child Medicaid coverage would build the outcome into the group definition. Castillo, Hill, and Hertz (2024) validate the logical-imputation algorithm against a survey with a well-regarded status measure and find that Medicaid participation is the single variable posing the greatest challenge to its accuracy. Every benefit-use, insurance, and income variable is therefore excluded from the status model.

The classification proceeds in three steps. First, deterministic lawful screens restricted to those that do not require benefit data: arrival before 1980; refugee and humanitarian origin crossed with arrival cohort; and occupations requiring licensure, citizenship, or federal clearance. Three screens from the published algorithms are deliberately not applied: public benefit receipt, for the reason above; “another family member is a U.S. citizen,” which is degenerate here because the frame is defined as parents of citizen

children; and “spouse is a U.S. citizen,” which does not itself confer status and enters only as a model predictor. Second, the residual noncitizens are ranked by a naturalization-propensity model, an L2-regularized weighted logistic regression fit on foreign-born adults with at least five years of tenure in survey years 2014-2018 only, with a weighted AUC of 0.866. Third, the ranking is calibrated to an external total by a single scalar. The scalar is chosen so that the weighted share of noncitizen adults classified as imputed likely undocumented in 2014-2018 matches the Pew Research Center estimate of 10.5 million unauthorized immigrants in 2017 divided by the ACS-weighted noncitizen population in 2017 (Pew Research Center, 2019, 2025), giving a target share of 0.4675 and a calibration constant of 0.7316. Both the model coefficients and the calibration constant are frozen on pre-2019 data, and every predictor is predetermined relative to the policy break. Twenty status draws are taken at the parent level and combined by Rubin’s rules. Variant A uses the screens plus the calibrated model; Variant B uses the screens only, treating every unscreened noncitizen parent as possibly undocumented, which is a maximal-exposure bound rather than an estimate.

Table I1. ACS Children Change in Medicaid/CHIP coverage over time, imputed parental legal status

Variant	Group (reference: U.S.-born parents)	Estimate (pp)	SE	95% CI	p
A: screens plus calibrated model	Long-naturalized parent by post	+0.91	0.80	[-0.70, +2.52]	0.262
A: screens plus calibrated model	Imputed likely lawfully present noncitizen parent by post	-1.05	0.79	[-2.65, +0.55]	0.191
A: screens plus calibrated model	Imputed likely undocumented parent by post	-2.15	0.66	[-3.49, -0.81]	0.002
B: screens only	Long-naturalized parent by post	+0.91	0.80	[-0.70, +2.53]	0.260
B: screens only	Imputed likely lawfully present noncitizen parent by post	-0.68	1.22	[-3.12, +1.77]	0.581
B: screens only	Imputed likely undocumented parent by post	-1.82	0.68	[-3.18, -0.45]	0.010

Notes: Each variant is a single multi-group weighted least squares regression on U.S.-born citizen children below 200% FPL, 2014-2022 (N = 1,750,526), with state and year fixed effects, the Section 3.5 controls, and standard errors clustered on state. This appendix uses the U.S.-born child frame rather than the all-citizen frame of the primary sample, so its estimates are not directly comparable in level to Table 3. Variant A combines 20 status draws by Rubin’s rules; Variant B is deterministic. The gap between variants is 0.33 points on the imputed-undocumented term and 0.38 points on the imputed-lawfully-present term, which is the measure of how much of the gradient is model-driven. Variant B assigns every noncitizen parent who does not pass a narrow screen to the undocumented group, so its undocumented coefficient is close to the pooled noncitizen-parent estimate by construction. No benefit, insurance, or income variable enters the status model. Author’s calculations from American Community Survey microdata (IPUMS USA, 2014-2022).

Table I2. Share-stability diagnostic for the imputed status classification

Year	Imputed likely undocumented share, noncitizen adults	Screened-lawful share	Mean probability at least one parent imputed undocumented
2014	0.458	0.146	0.639
2015	0.468	0.136	0.643
2016	0.470	0.137	0.648
2017	0.469	0.138	0.642

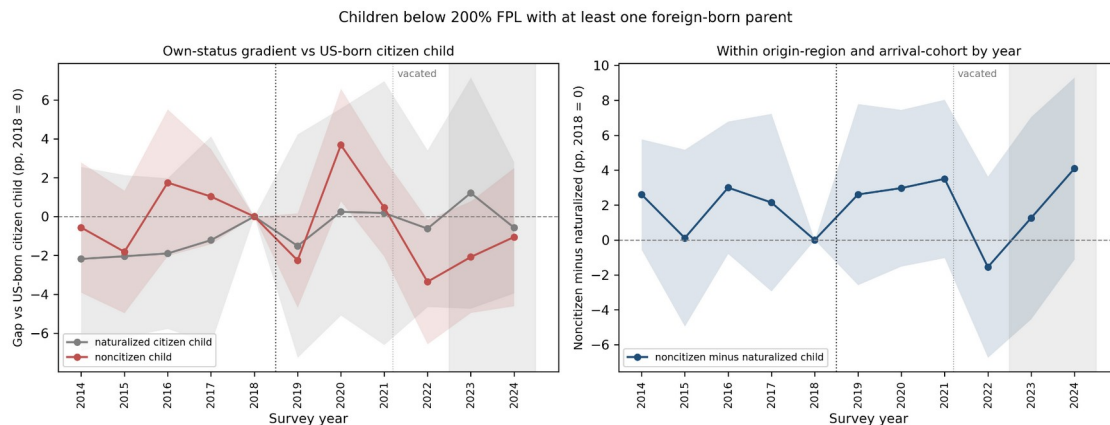
2018	0.473	0.135	0.640
2019	0.478	0.128	0.644
2020	0.474	0.125	0.631
2021	0.485	0.117	0.640
2022	0.492	0.112	0.644

Notes: Because the model coefficients and the calibration constant are frozen on 2014-2018 data and every predictor is predetermined relative to the policy break, the classified shares should move only with compositional change in the noncitizen population. The national noncitizen-adult share drifts from a pre-2019 mean of 0.4675 to a 2019-onward mean of 0.4822, and the drift comes from the two cohort-based screens ageing out of the working-age population: pre-1980 arrivals fall from 0.068 to 0.043 and refugee-origin cells from 0.057 to 0.041, while the licensed-occupation screen is flat. The series that governs group assignment in the regression is the last column, which moves between 0.631 and 0.648 across all nine years with a pre-post difference of -0.3 points and no trend. Author's calculations from American Community Survey microdata (IPUMS USA, 2014-2022).

Limitations of this exercise. No legal status is observed; the procedure ranks and calibrates rather than classifying. The ranking device is a naturalization-propensity model, so it will misrank lawfully present noncitizens who are ineligible or slow to naturalize, including recent lawful permanent residents, temporary visa holders, and TPS and DACA recipients, toward higher risk. That mixes the two noncitizen groups and attenuates the gradient instead of manufacturing it. The calibration target is national and all-ages, applied to noncitizen adults, while parents of citizen children below 200% FPL are not a random draw from that population. Variant B is a bound and not an estimate. The methodological reference points for the residual and multiple-imputation approaches are Borjas (2017), Van Hook et al. (2015), Capps, Bachmeier, and Van Hook (2018), and Castillo, Hill, and Hertz (2024).

Appendix J. The child's own citizenship status

Figure J1. Event study, child's own citizenship status within immigrant-parent families



Notes: Event-study coefficients for noncitizen and naturalized children relative to U.S.-born citizen children within families with at least one foreign-born parent, reference year 2018, below 200% FPL, with 95% confidence intervals. The dotted line marks the September 2018 proposal, and the lighter dotted line marks the March 2021 vacatur. Author's calculations from American Community Survey microdata (IPUMS USA, 2014-2024).

Table J1. ACS Children Change in Medicaid/CHIP coverage over time, child's own citizenship status

Specification	Group	Estimate (pp)	SE	p	MDE (80%)	N
Within foreign-born-parent families, state and year FE	Noncitizen child	-0.70	0.75	0.355	2.11	554,562

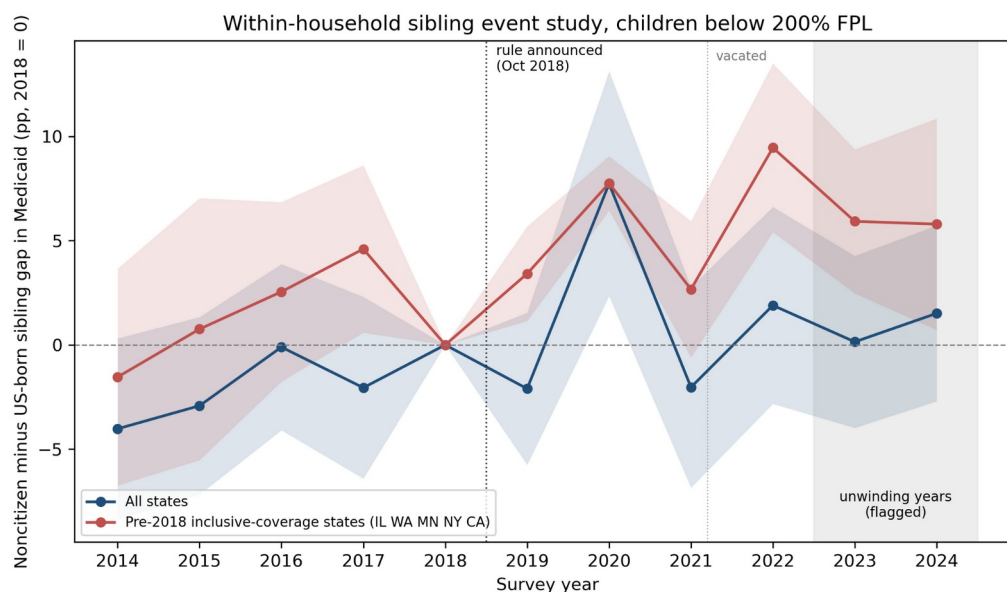
Holding parental exposure fixed (noncitizen parent present)	Noncitizen child	-0.17	0.69	0.811	1.94	386,369
Foreign-born children only, origin-by-year and cohort-by-year FE	Noncitizen minus naturalized child	+0.24	1.55	0.876	4.33	77,824

Notes: The reference group is U.S.-born citizen children in the first two rows and naturalized children in the third, where the U.S.-born reference is absorbed by the origin-by-year and arrival-cohort-by-year fixed effects. A sensitivity restricting the foreign-born groups to children who arrived before 2017 gives +5.24 points; that restriction makes the foreign-born groups a closed cohort accumulating U.S. tenure past the five-year bar while the U.S.-born reference is replenished, which moves the coefficient upward mechanically, so it is recorded and not interpreted. Pooled window 2014-2022, weighted least squares, standard errors clustered on state. Author's calculations from American Community Survey microdata (IPUMS USA, 2014-2022).

The child's own citizenship status adds nothing on top of the family-level contrast, which is what the rule itself predicts, since Medicaid received by a noncitizen under age 21 was excluded from the countable benefit definition.

Appendix K. Within-household sibling comparison

Figure K1. Event study, noncitizen child relative to U.S.-born sibling



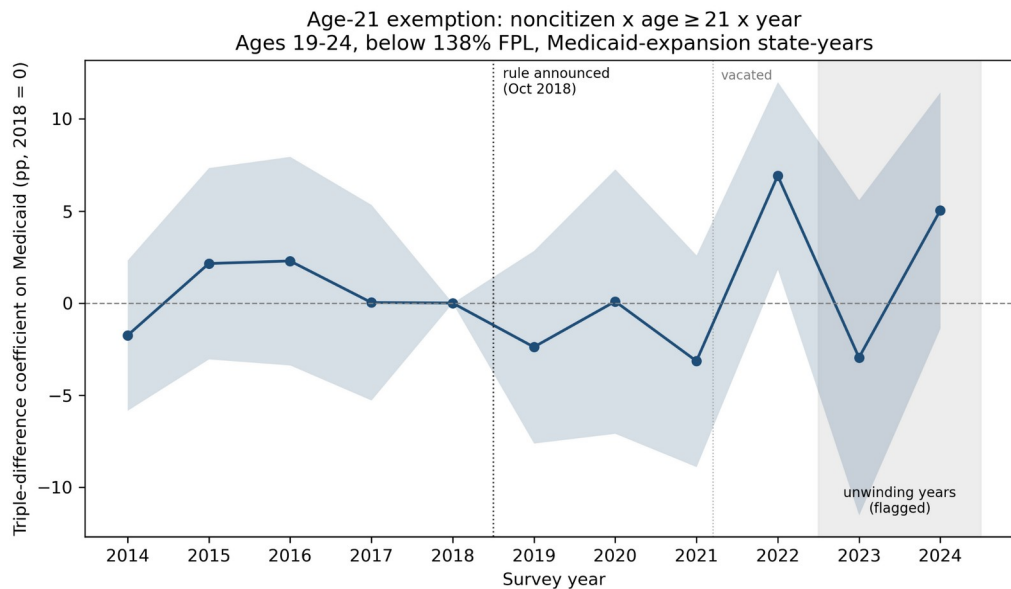
Notes: Event-study coefficients on the noncitizen-child indicator within households containing both a U.S.-born citizen child and a noncitizen child sharing resident parents, with household fixed effects and child age-by-year effects, reference year 2018. The dotted line marks the September 2018 proposal, and the lighter dotted line marks the March 2021 vacatur. Author's calculations from American Community Survey microdata (IPUMS USA, 2014-2024).

Within the 17,307 below-200% FPL households containing both a U.S.-born citizen child and a noncitizen child sharing resident parents (51,200 children over 2014-2022), the noncitizen-child by post coefficient is +2.92 points (SE 1.33, $p = 0.033$, minimum detectable effect 3.73 points). It falls to +1.57 points (SE 1.45, $p = 0.28$) once the 2020-2021 continuous-enrollment years are dropped, so the positive point estimate is a public health emergency artifact and not a finding. The joint pre-trend test rejects ($p = 0.035$). This design cannot recover the paper's question by construction, because household fixed effects

remove exactly the family-level response the primary design measures, leaving only whether withdrawal was aimed differentially at the child whose own status was at stake. Since the 2019 rule excluded Medicaid received by a noncitizen under 21, and several states extended child coverage regardless of status, a null or positive contrast is what the rule itself predicts.

Appendix L. Age-21 exemption

Figure L1. Event study, age-21 exemption triple difference



Notes: Event-study coefficients on the noncitizen by age-21-or-older interaction among 19- to 24-year-olds below 138% FPL in Medicaid-expansion state-years, reference year 2018, with 95% confidence intervals. The dotted line marks the September 2018 proposal, and the lighter dotted line marks the March 2021 vacatur. Author's calculations from American Community Survey microdata (IPUMS USA, 2014-2024).

The 2019 rule excluded Medicaid received by a noncitizen under 21 while Medicaid received from age 21 could count, which suggests a discontinuity test. Among 233,883 people aged 19 to 24 below 138% FPL in Medicaid-expansion state-years, the noncitizen by age-21-or-older by post coefficient is -0.09 points (SE 1.50, $p = 0.95$), with a joint pre-trend test that passes comfortably ($p = 0.325$). The null is stable across every cut: -0.52 points on uninsurance, +1.66 dropping 2020-2021, +0.25 on the balanced always-expansion panel, +0.22 excluding California's 2020 Medi-Cal expansion for undocumented 19- to 25-year-olds, and -1.73 on the tightest age-20-versus-21 window. Power is the limitation. The minimum detectable effect is 4.20 points, roughly twice the magnitude the paper reports, so this design constrains only large effects. Age 21 also coincides with other coverage transitions, the ACS records age at interview instead of at benefit receipt, and many noncitizens in this age band are barred from federally funded Medicaid regardless of the rule.

Appendix M. Predicted eligibility

Table M1 replaces the blunt income screen with the child's own state, year, and age-band Medicaid or CHIP income threshold.

Table M1. ACS Children Change in Medicaid/CHIP coverage over time, predicted eligibility

Specification	Estimate (pp)	SE	p	N
DiD, children below their applicable threshold	-2.20	0.43	<0.001	2,384,713

DiD, children above their applicable threshold	+1.71	0.42	<0.001	3,048,240
DiD, blunt below-200% FPL screen, same sample rules	-1.91	0.45	<0.001	1,804,558
Triple difference, raw	-4.14	0.42	<0.001	5,432,953
Triple difference, income-bin by group by period FE (preferred)	+0.02	0.89	0.978	5,432,953
Triple difference, within 50 FPL points of the threshold	-1.63	0.88	0.069	954,053

Notes: Predicted eligibility merges a state-by-year-by-age-band panel of children's Medicaid and CHIP upper income thresholds onto each child record and compares it with the household income-to-poverty ratio. Triple-difference rows include state-by-year fixed effects and all lower-order terms; the preferred row adds fixed effects for 10-FPL-point income bins interacted with group and period, so the interaction is identified only from where the cutoff falls across states and age bands at a given measured income. That row is stable at -0.09 under 25-point bins and +0.02 using age-band threshold variation alone, with an 80%-power minimum detectable effect of 2.50 points. Author's calculations from American Community Survey microdata (IPUMS USA, 2014-2022) merged to state children's eligibility thresholds.

The eligibility merge backs the low-income sample restriction, and it does not deliver an independent eligibility-based discontinuity. The decline is concentrated below the applicable threshold at -2.20 points, close to the -1.91 the blunt below-200% FPL screen gives on the same sample rules, so the result is not an artifact of the 200% cut.

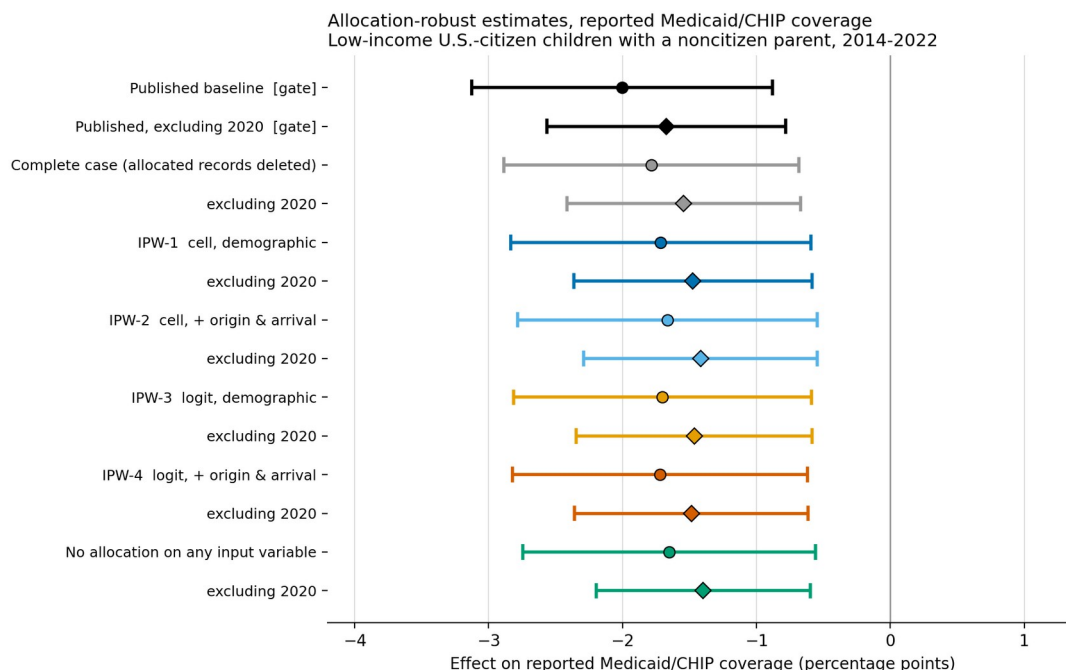
Once income position is held fixed, the triple difference sits near zero. The raw triple difference of -4.14 points exceeds the below-threshold decline itself because the above-threshold group moved the other way, gaining +1.71 points, which no public charge mechanism predicts and which signals that the eligibility split partly captures differential income-position drift: the predicted-eligible share fell 11.9 points among mixed-status children between 2014 and 2022 against 7.2 points among citizen-parent children. Once 10-FPL-point income bins interacted with group and period hold income position fixed, the triple difference is +0.02 points (SE 0.89, $p = 0.978$), stable under alternative bin widths and under age-band threshold variation alone, and the narrow-bandwidth version attenuates to -1.63 points ($p = 0.069$). The below-threshold event study also has significantly positive 2014 and 2015 coefficients. The data do not identify an additional discontinuity at the estimated eligibility threshold.

Appendix N. Exposure allocation: response weighting and reclassification scenarios

Parent citizenship defines exposure, and after 2018 it is increasingly supplied by the Census imputation model: the weighted share of families with an allocated parent-citizenship response roughly doubles across the pre and post split, from 7.04% to 14.62% for mixed-status families against 5.54% to 9.81% for citizen-parent families, a differential of 3.32 points (3.43 points, SE 0.41, with the regression controls). That raises two problems, who reports and what the model assigns, and this appendix bounds both.

Selection into reported responses. The complete-case sample of reported responses carries no model-determined exposure assignment, and inverse-probability weighting returns it to the full estimation sample under the assumption that response is ignorable given the covariates. All four response models give the same answer: the headline moves from -2.00 to -1.67 points and the companion excluding-2020 estimate from -1.67 to -1.42, an attenuation of roughly 14 to 25%, with every p -value at or below 0.0043. The weighting restores covariate balance at a cost of about 9% of effective sample size. Figure N1 plots the estimates, and Tables N1 and N2 report them.

Figure N1. Allocation-robust estimates, reported Medicaid/CHIP coverage, 2014-2022



Diamonds mark the variant that drops the 2020 experimental ACS. IPW rows reweight children whose parents' citizenship was reported rather than allocated back to the full estimation sample. Whiskers are 95% confidence intervals from standard errors clustered on state (51 clusters).

Notes: Point estimates and 95% confidence intervals for the baseline, complete-case, and inverse-probability-weighted specifications reported in Tables N1 and N2; diamonds mark the variant that drops the experimental 2020 ACS. Author's calculations (IPUMS USA, 2014-2022).

Potential exposure misclassification. Weighting does not correct error in what the imputation model assigns, which can only be bounded under directional assumptions. The sign survives every wholesale reclassification in Table N2 except one: sending every post-2018 allocated record to the comparison group gives -0.40 points ($p = 0.418$). Allocated records report lower coverage than reported records by a nearly constant margin in every group-by-period cell, which argues against error sorted on the child's coverage status. However, whether the imputation errs in a way correlated with both group and outcome cannot be settled with these data.

Table N1. ACS Children Change in Medicaid/CHIP coverage over time, allocated parent citizenship

Specification	Estimate (pp)	SE	p	Excluding 2020
Baseline	-2.00	0.56	<0.001	-1.67
Complete case: parent citizenship reported	-1.78	0.55	0.002	-1.54
Inverse-probability weighted, cell propensity, adding origin and arrival	-1.67	0.56	0.004	-1.42
All post-2018 allocated records assigned to the comparison group	-0.40	0.49	0.418	-0.19

Notes: A record is allocated when at least one resident parent's citizenship item was imputed by the Census Bureau. The weighted row reweights children whose parents' citizenship was reported by the inverse of the estimated probability of a reported response; the response models use only covariates observed for every child and never the exposure indicator, since exposure is what is missing when a parent's citizenship is allocated. The final row moves every post-2018 allocated record into the comparison group, which bounds a complete one-directional failure of the allocation model. Table N2 reports three further weighting models and the remaining scenarios. Weighted least squares, state and year fixed effects, Section 3.5 controls, standard errors clustered on state. Author's calculations (IPUMS USA, 2014-2022).

Table N2. ACS Children Change in Medicaid/CHIP coverage over time, allocation weighting models and reclassification scenarios

Specification	Estimate (pp)	SE	p	Excluding 2020
Inverse-probability weighted, cell propensity, demographic	-1.71	0.56	0.003	-1.47
Inverse-probability weighted, logit, demographic	-1.70	0.55	0.003	-1.46
Inverse-probability weighted, logit, adding origin and arrival	-1.72	0.55	0.003	-1.48
Complete case: no allocation on any input variable	-1.65	0.54	0.004	-1.40
All allocated records assigned to the comparison group	-1.31	0.49	0.010	-1.09
All allocated records assigned to the exposed group	-3.50	0.52	<0.001	-3.21
All post-2018 allocated records assigned to the exposed group	-6.68	0.51	<0.001	-6.41

Notes: A record is allocated when at least one resident parent's citizenship item was imputed by the Census Bureau. The weighted rows reweight children whose parents' citizenship was reported by the inverse of the estimated probability of a reported response, using models containing only covariates observed for every child. The reclassification rows move whole blocks of allocated records into one group and re-estimate; they are bounding exercises, and none is put forward as a plausible scenario. Weighted least squares, state and year fixed effects, Section 3.5 controls, standard errors clustered on state. Author's calculations from American Community Survey microdata (IPUMS USA, 2014-2022).

Weighting diagnostics. Weighting to the probability of a reported response restores covariate balance against the full estimation sample. The maximum absolute standardized difference falls from 0.031 to at most 0.0098, and the maximum by-year gap in the exposed share falls from 1.19 to 0.28 points. The cost is about 9% of effective sample size. The weighting leaves the joint 2014-2017 lead test where it was, at $p = 0.161$ against 0.178 unweighted.

Reclassification scenarios. Under an error model in which the group of a misassigned record is redrawn from that year's reported-record mix, the estimate never reaches zero: it is -1.86 points ($p = 0.0001$) even at a 100% error rate. Under an adversarial model in which the misassigned records are selected on the child's coverage status, the result breaks at 5.1% of allocated records. Allowing every allocated record's group to be chosen freely bounds the estimate at [-23.20, +16.62] points. In short, the data alone place almost no restriction on this problem.

Appendix O. CHIS fear-based benefit avoidance estimates

Table O1 reports the estimates plotted in Figure 1.

Table O1. CHIS Adults Fear-based benefit avoidance, adults living with children vs all adults, 2019-2024
Panel A. Ever decided not to apply for a non-cash benefit over immigration concerns

Documentation group	Adults living with children: share (SE) [95% CI], n	All adults: share (SE) [95% CI], n
All foreign-born groups	25.5% (0.7) [24.1, 27.0], 9,965	17.7% (0.3) [17.0, 18.3], 33,464
Naturalized citizen	13.6% (0.6) [12.5, 14.8], 6,032	9.7% (0.3) [9.1, 10.2], 24,085

Green-card holder	27.6% (1.5) [24.7, 30.5], 2,122	22.1% (0.8) [20.6, 23.6], 5,499
No green card	45.0% (1.6) [41.7, 48.2], 1,811	37.9% (1.0) [35.9, 40.0], 3,880

Panel B. Among those who ever avoided, avoided in the past 12 months

Documentation group	Adults living with children: share (SE) [95% CI], n	All adults: share (SE) [95% CI], n
All foreign-born groups	50.6% (1.3) [48.0, 53.2], 1,988	45.5% (1.0) [43.5, 47.5], 4,198
Naturalized citizen	41.0% (2.4) [36.3, 45.7], 666	34.0% (1.7) [30.7, 37.3], 1,737
Green-card holder	53.9% (2.9) [48.2, 59.5], 500	48.9% (2.1) [44.8, 53.0], 1,076
No green card	54.2% (2.2) [49.9, 58.4], 822	52.5% (1.6) [49.3, 55.7], 1,385

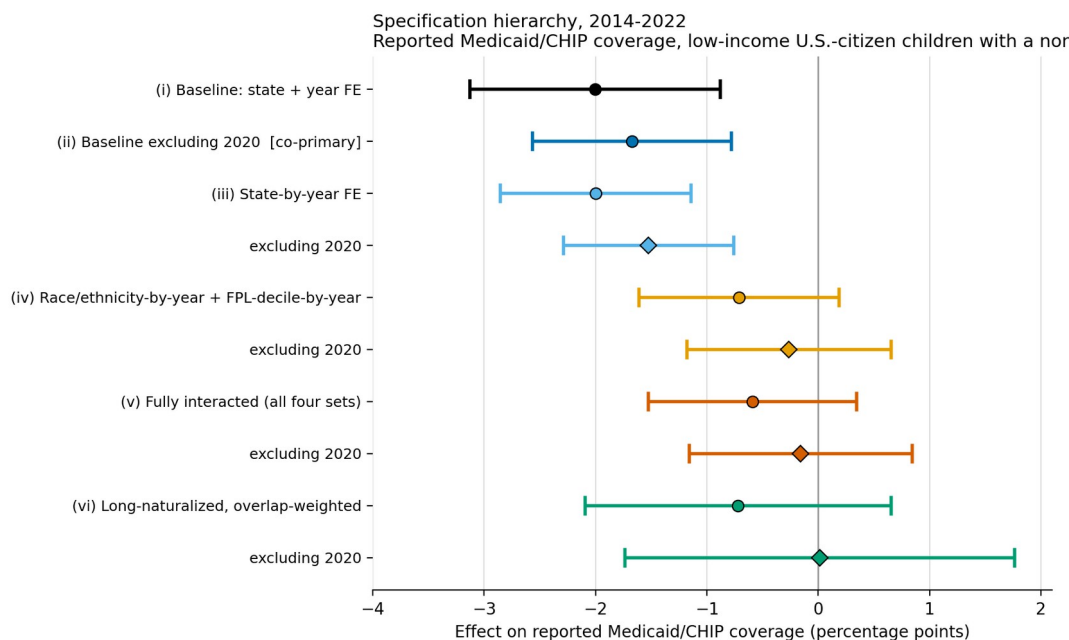
Notes: Survey-weighted shares with jackknife replicate-weight standard errors in parentheses, 95% confidence intervals in brackets, and unweighted denominators (n), pooled over the 2019-2024 CHIS adult files, the years in which both items are fielded. Adults living with children are respondents in households with at least one child under 18. Both items are asked only of foreign-born respondents, Panel B is conditional on having ever avoided, and the Panel A item is retrospective and unbounded in time. Author's calculations from restricted-use CHIS data accessed through the UCLA Data Access Center.

Appendix P. Specification hierarchy estimates

This appendix reports the primary estimate as a hierarchy of nine specifications, grouped by what each one holds fixed. Five use the citizen-parent comparison group: (i) baseline state and year fixed effects; (ii) the same model excluding the experimental 2020 ACS; (iii) state-by-year fixed effects, which absorb every state-specific annual shock; (iv) race and ethnicity-by-year plus FPL-decile-by-year fixed effects; and (v) all four characteristic-by-year interaction sets together. One, (vi), switches to the long-naturalized comparison group with overlap weights. Three replace the contemporaneous income screen with a fixed pre-period geography rule, classifying each area once from pre-2019 data so that no child's inclusion can respond to the policy: (vii) all children in PUMAs whose pre-2019 share of citizen children below 200% FPL is above the median, (viii) the same design in top-tercile PUMAs, and (ix) the same design at the state level, where boundaries are stable across the window.

These specifications trade off against each other in ways the data cannot resolve. The baseline preserves any community-wide immigration-policy spillover, and it is open to unrelated shocks specific to Hispanic or low-income families; the richer models remove those shocks, and they may also absorb spillovers reaching Hispanic citizen-parent families. Within Panel A of Table P1 the estimates run from -2.00 to -0.59 points, and from -1.67 to -0.16 excluding the experimental 2020 ACS; rows (i), (ii), and (iii) are significant at the 5% level and rows (iv) and (v) are not. Panel B gives -0.72 points and +0.01 excluding 2020, on a different population. Panel C runs from -2.41 to -1.92, and it carries a pre-trend weakness of its own: the joint 2014-2017 lead tests reject at $p < 0.001$ for the PUMA classification and $p = 0.001$ for the state classification, against $p = 0.178$ for the primary design, and the Rambachan and Roth (2023) breakdown values run 0.30 and 0.40 against 0.25. The fixed-geography designs also leave row (iv) where it was, since income-position-by-year fixed effects inside the fixed-geography sample give -0.08 points ($p = 0.921$). Geography is not what drives the attenuation in Panel A: state-by-year fixed effects leave the estimate at the baseline level with a tighter interval.

Figure P1. Specification hierarchy, reported Medicaid/CHIP coverage, 2014-2022



Diamonds mark the variant that drops the 2020 experimental ACS. Whiskers are 95% confidence intervals from standard errors clustered on state (51 clusters). Row (vi) uses a different comparison group (families in which every immigrant parent naturalised by 2013) and estimates an overlap-weighted ATO rather than an ATT.

Notes: Point estimates and 95% confidence intervals for the six rows of Panels A and B of Table P1; diamonds mark the variant that drops the experimental 2020 ACS. Row (vi) uses a different comparison group and estimand. Author's calculations (IPUMS USA, 2014-2022).

Table P1. ACS Children Change in Medicaid/CHIP coverage over time, specification hierarchy

Panel A. Same comparison group: citizen-parent families below 200% FPL

Specification	Interpretive role	Estimate (pp)	SE	95% CI	p	N
(i) Baseline: state and year FE	Conventional, minimally adjusted group-change contrast	-2.00	0.56	[-3.13, -0.88]	<0.001	1,841,358
(ii) Baseline excluding 2020	Companion data-quality specification	-1.67	0.44	[-2.56, -0.78]	<0.001	1,697,491
(iii) State-by-year FE	Removes all state-specific annual shocks	-2.00	0.43	[-2.85, -1.14]	<0.001	1,841,358
(iii) State-by-year FE, excluding 2020		-1.52	0.38	[-2.29, -0.76]	<0.001	1,697,491
(iv) Race and ethnicity-by-year and FPL-decile-by-year FE	Principal group-specific confounding test	-0.71	0.45	[-1.61, +0.19]	0.118	1,841,358
(iv) Same, excluding 2020		-0.26	0.46	[-1.18, +0.65]	0.569	1,697,491
(v) Fully interacted: age-, race-, family-structure- and FPL-	Most demanding observed-characteristic trend adjustment	-0.59	0.47	[-1.52, +0.35]	0.211	1,841,328

decile-by-year FE						
(v) Same, excluding 2020		-0.16	0.50	[-1.16, +0.84]	0.753	1,697,463
Panel B. Alternative comparison group and estimand: long-naturalized-parent families						
Specification	Interpretive role	Estimate (pp)	SE	95% CI	p	N
(vi) Long-naturalized comparison, overlap-weighted	Immigrant-background cross-check	-0.72	0.68	[-2.09, +0.65]	0.298	394,553
(vi) Same, excluding 2020		+0.01	0.87	[-1.74, +1.76]	0.988	367,964
Panel C. Income-selection diagnostics: fixed pre-period geography, no contemporaneous income screen						
Specification	Interpretive role	Estimate (pp)	SE	95% CI	p	N
(vii) PUMAs above the median child-poverty share, all incomes	Income leaves the sample rule; fixed pre-period geography	-2.22	0.76	[-3.75, -0.68]	0.006	2,421,202
(vii) Same, excluding 2020		-1.92	0.64	[-3.21, -0.63]	0.005	2,190,442
(viii) Top-tercile PUMAs, all incomes	Same design, more concentrated geography	-2.02	0.73	[-3.49, -0.55]	0.008	1,569,474
(viii) Same, excluding 2020		-1.98	0.60	[-3.19, -0.78]	0.002	1,424,401
(ix) States above the median child-poverty share, all incomes	State-level version on stable boundaries	-2.41	0.62	[-3.69, -1.13]	<0.001	3,692,991
(ix) Same, excluding 2020		-1.92	0.59	[-3.14, -0.69]	0.003	3,384,203

Notes: The three panels do not estimate the same parameter, so no single range across them is reported. Panel A uses U.S.-citizen children below 200% FPL over 2014-2022 with the Section 3.5 controls, replaced by the corresponding year interaction where a characteristic enters interacted. Panel B estimates an overlap-weighted effect on the overlap population of families in which every immigrant parent naturalized and arrived by 2013. Panel C classifies each PUMA or state once from pooled 2014-2018 data by its weighted share of citizen children below 200% FPL and includes every child living there at any income. PUMA rows cover 2014-2021 because PUMA boundaries were redrawn for the 2020 census, and state rows cover 2014-2022 on 26 clusters. The Panel C designs fail their joint 2014-2017 lead tests: $p < 0.001$ for the PUMA classification of rows (vii) and (viii) and $p = 0.001$ for the state classification of row (ix), against $p = 0.178$ for the primary design. Author's calculations.

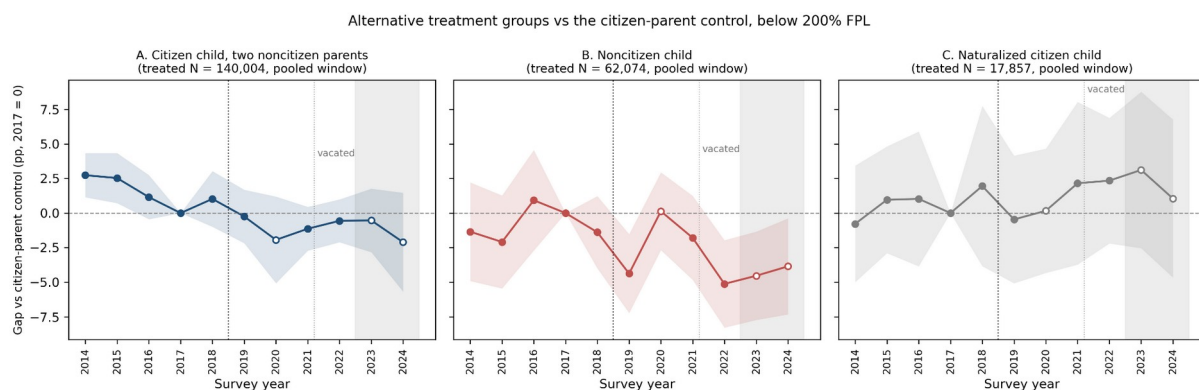
Appendix Q. Alternative treatment groups

The headline design fixes the treatment group as citizen children with at least one noncitizen resident parent. Table 10 re-estimates the pooled and phase models with that group replaced, in turn, by three alternatives, holding the comparison group at the paper’s control: citizen children whose identified resident parents are all citizens. This appendix records the design details and exclusion logic and reports the corresponding event studies. In each design, every other exposed group is removed from the estimation sample rather than left to sit in the comparison group. The first design takes U.S.-born citizen children with two identified resident parents, both noncitizens, the most exposed parental configuration in Table 8; citizen children with exactly one noncitizen parent and all foreign-born children are excluded, and both treatment and comparison groups are restricted to two-resident-parent families, since the treatment definition mechanically requires two resident parents and an unrestricted comparison group would fold a family-structure difference into the contrast. Relaxing that restriction moves the estimate from -2.49 to -1.98 points (SE 0.70, $p = 0.007$). The second design takes noncitizen children, pooling lawful permanent residents and undocumented children as in Figure 4, and excludes naturalized children and every citizen child with a noncitizen parent. The third takes naturalized citizen children and excludes noncitizen children and every non-naturalized citizen child with a noncitizen parent.

These designs do not all estimate the spillover parameter the paper is built around. The first does: the children are U.S.-born citizens whose own eligibility rules did not change, so it is the headline estimand at maximal parental exposure. The second and third place children who are themselves exposed, or whose own status changed through naturalization, in the treatment group. That also distinguishes this exercise from Appendix J, which compares children by their own status within immigrant families; here each group is contrasted against the citizen-parent control. Noncitizen children additionally face different eligibility rules, including the five-year bar and state coverage regardless of status, so their estimate mixes chilling with direct exposure and cannot be read as a chilling effect alone.

Table 10 in the main text reports the pooled and phase estimates and the lead-test results. Figure Q1 reports the event studies underlying its joint pre-trend column. One level difference matters when reading the noncitizen-child panel: that group’s coverage sits near 52% throughout against 71% for the control, so the two series differ far more in level than the headline comparison does.

Figure Q1. Event studies, alternative treatment groups vs the citizen-parent control, below 200% FPL



Notes: Event-study coefficients on the treatment-group-by-year interactions, estimated over 2014-2024 with 2017 as the omitted reference and 2018 carrying its own transition coefficient, state and year fixed effects, the Section 3.5 controls, and standard errors clustered on state; shaded bands are 95% confidence intervals. Each panel replaces the mixed-status treatment group with the named group, contrasted against citizen children whose identified resident parents are all citizens, with the other exposed groups excluded from the sample as described in the Table 10 notes; panel titles give each group’s pooled-window treated child-years. The dotted line marks the September 2018 proposal, and the lighter dotted line marks the March 2021 vacatur. Hollow markers denote the experimental 2020 ACS and the 2023-2024 unwinding years, and the shaded region covers the unwinding years. The unwinding years are excluded from every pooled estimate; the 2020 file is included in the pooled estimates and dropped in the excluding-2020 column. Author’s calculations from American Community Survey microdata (IPUMS USA, 2014-2024).

